

Final TIMES Project

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Introduction

The present study analyzes what may be the future development of the Netherlands' electricity system, including the potential impact of a diverse set of policy instruments. The analysis uses the TIMES energy system model, according to a technology-optimization approach. This model is implemented by the VEDA platform, allowing complex long-term energy planning scenarios to be managed and built.

Seven distinct policies are evaluated since they range from carbon pricing mechanisms coupled with emission caps to investment mandates along with renewable energy subsidies each representing different strategies within the broader energy transition framework. The central aim of the project involves judging how these tools might help climate and energy objectives in the long run. The assessment will consider individual instruments and combinations throughout 2025–2050.

These policies reflect real-world planned priorities such as an increase in the share of renewable energy, reduction of carbon intensity, and continuous availability of carbon-free electricity. The effects on operational performance as well as system costs were examined, also technology uptake including greenhouse gas emissions coupled with electricity generation mix through scenario-based modeling.

Objective

The objective of this project is to assess the impact of 7 different policies in the power sector from 2025 to 2050. These instruments are designed to show the various approaches to decarbonization, through carbon taxation, renewable energy mandates, emission caps, among other strategies.

By modelling each policy individually, and for further understanding, some of the policies were combined. The project uses TIMES through the VEDA platform seeking to assess their effectiveness when achieving key energy and climate goals. A specific focus is analysis of changes in electricity generation mix, GHG emissions, renewable energy integration, technology deployment, and grid flexibility, as well as system costs.

To have a deeper understanding of the effect of policies in decarbonization strategies, the project also explores a combination of the most effective policies, specifically subsidies, recycling of carbon tax revenues, CO₂ emissions cap and carbon tax. This mixture of policies is done to assess whether this combination is adequate to accelerate the transition to reach full decarbonization. The aim is to identify which policies are the most effective to reach the climate targets.

System Topology

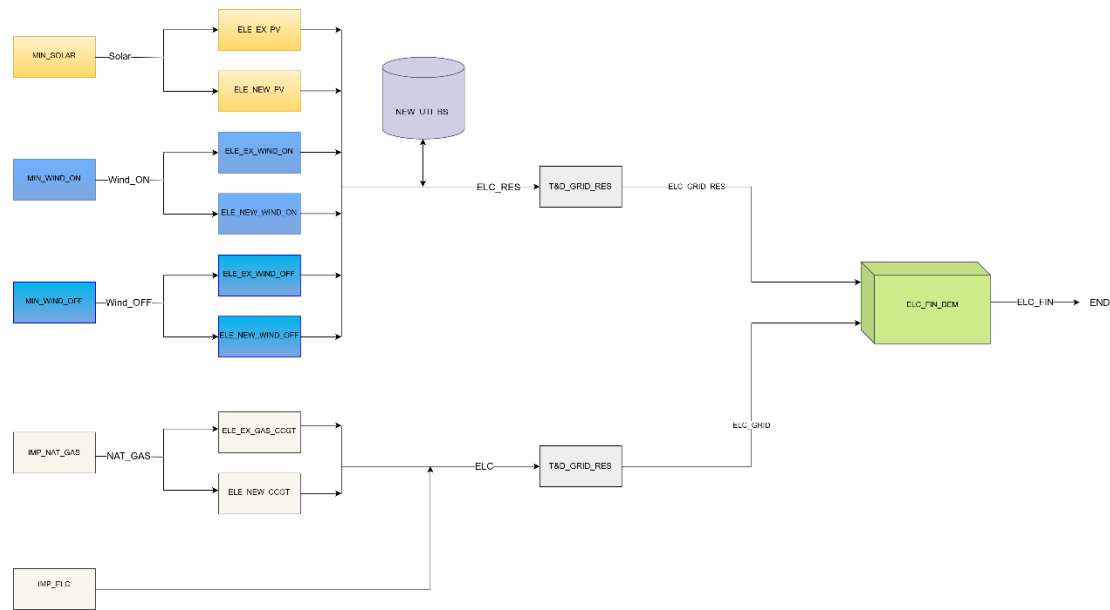


Figure 1: System Topology

This figure indicates the topology of the energy system modelled in the Netherlands. It consists of energy generation, transmission, storage, and final demand. The system includes various electricity generation sources such as solar photovoltaic, onshore wind, offshore wind, and natural gas power plants. These sources represent both existing capacities and potential new installations that could be added in the future.

In addition to local generation, the system allows for electricity imports and natural gas imports to ensure supply security. A utility-scale battery storage system is also included, which can store excess electricity and release it when needed, helping to balance supply and demand.

All generated and imported electricity is gathered into a central electricity pool. From there, electricity is transmitted and distributed through the transmission and distribution grid to reach end users.

The final electricity demand represents the total consumption by households, industries, and other consumers.

Methodology

All policy scenarios were modeled using the TIMES energy system model. TIMES gives raw outputs in terms of energy flows, capacities, and system costs. The TIMES outputs were then processed in Excel to calculate the relevant energy indicators. The variables and calculations for all cases were as follows:

Electricity Generation Mix

- Process: ELE_EX_GAS_CCGT, ELE_NEW_CCGT, ELE_PV, ELE_WIND_ON, and ELE_WIND_OFF
- Attribute: VAR_FOut (in PJ).

For the conversion to TWh the following formula was used:

$$\text{Electricity generation [TWh]} = \text{VAR}_{\text{Fout}} \cdot 0.27778$$

Overall GHG Emissions

- Process: ELE_EX_GAS_CCGT and ELE_NEW_CCGT.
- Attribute: VAR_Comnet (in kt CO₂).

For the conversion in Mt CO₂:

$$\text{GHG emissions [Mt CO}_2\text{]} = \frac{\text{VAR}_{\text{Comnet}}}{1000}$$

Carbon Intensity of Electricity Generation

$$\text{Carbon intensity [kg CO}_2\text{ / MWh]} = \frac{\text{Total CO}_2\text{ emissions [kt]} \times 1,000}{\text{Total Electricity Generation [TWh]} \times 1000}$$

This results in the carbon intensity for each policy from year 2025 to 2050.

The import of electricity has also been considered in the carbon intensity of electricity generation, as its share is relatively small.

New Electrical Capacities Added

- Process: ELE_NEW_CCGT and ELE_WIND_ON
- Attribute: VAR_Ncap (GW)

This captures the annual new installed capacity.

Full Load Hours of CCGT

- Process: ELE_EX_CCGT, ELE_NEW_CCGT
- Attributes:
 - VAR_Cap (GW): installed capacity
 - VAR_FOut (PJ): Commodity used ELC, which gives the annual activity.
- Calculation:

$$\text{Annual activity [GWh]} = \frac{\text{VAR}_{\text{FOut total}} \times 1,000}{3.6}$$

$$\text{FLH [h/yr]} = \frac{\text{Annual activity [GWh]}}{\text{Total VAR}_{\text{Cap}}[\text{GW}]}$$

Renewable Energy Curtailment

- Process: ELE_EX_PV, ELE_EX_WIND_OFF, ELE_EX_WIND_ON,
- Commodity: ELC, RES
- Attributes: VAR_FOut
- CAP2ACT: 31.536
- Average Power factor: 0.12 (PV), 0.49 (WIND OFF), 0.3 (WIND ON)
- Calculation:

$$\text{potential} = \frac{\text{CAP2ACT} * \text{VAR}_{\text{CAP}} * \text{Average power factor}}{3.6}$$

$$\text{actual} = \frac{\text{VAR}_{\text{Fout}}}{3.6}$$

$$\text{Curtailment [KWh]} = \text{Potential} - \text{Actual}$$

$$\text{Curtailment (\%)} = 1 - \frac{\text{Potential}}{\text{Actual}}$$

Renewable Share

- **RES electricity delivered to the grid:**
 - Commodity: ELC
 - Process: T&D_GRID_RES
 - Attribute: Val_Flo
- **Final electricity demand:**
 - Commodity: ELC_RES
 - Process: T&D_GRID_RES
 - Attribute: Val_Flo
- Calculation:

$$\text{RES Share (\%)} = \frac{\text{T\&D_GRID}}{\text{T\&D_GRID_RES}}$$

Key Constraints Considered for each Policy

0. **Baseline** - without any energy policy;
1. **Carbon Tax** – introducing a CO₂ tax to internalize the cost of emissions, the value is 0.07, 0.1, 0.2 0.3 MEur/kton for year 2023, 2030, 2040, 2050 respectively;
2. **Cap on CO₂ Emissions** – imposing an upper limit of 16000 kton on annual total emissions;
3. **Investment Mandate for Specific Technologies** – enforcing minimum capacity additions for technologies such as offshore wind; in this study, a minimum capacity is mandated for offshore wind, 1 GW in 2025 and 5 GW every 5 years from 2030 to 2050.
4. **Capital Subsidies** – reducing CAPEX for new offshore wind technology by introducing a subsidy of 2000 euro/kW.

5. **Recycling Carbon Tax Revenues** – combining CO₂ taxation with subsidies for renewable electricity consumption;
6. **RES Share in Annual Electricity Consumption** – ensuring that 50% of electricity consumption comes from RES annually;
7. **24/7 Carbon-Free Energy** – targeting 50–90% carbon-free electricity in every slice.

Result Analysis of Independent Policy

0. Baseline

0.1. Electricity Generation Mix [TWh]

Baseline (TWh)	2025	2030	2035	2040	2045	2050
CCGT	42.472	67.561	112.60	159.91	212.41	240.90
PV	20.794	16.300	10.967	5.483	0.000	0.000
Wind Off	12.232	9.394	6.261	3.131	0.000	0.000
Wind ON	47.308	61.684	62.053	58.417	50.736	57.542

Table 1: Electricity Generation Mix - Baseline



Figure 2: Electricity Generation Mix - Baseline

In the baseline case, the electricity generation mix becomes less sustainable over the years. Electricity production from Combined Cycle Gas Turbines (CCGT) continues to increase, while the share of Renewable Energy Sources (RES) declines. By 2050, most of the energy will be supplied by CCGT. This indicates that without the introduction of supportive policies or interventions, the country's energy system is unlikely to develop along a sustainable pathway spontaneously.

0.2. Overall GHG Emissions

Baseline (Mtons)	2025	2030	2035	2040	2045	2050
CO2 emission	15.017	23.888	39.816	48.714	64.706	73.385

Table 2: Overall CO2 emissions - Baseline

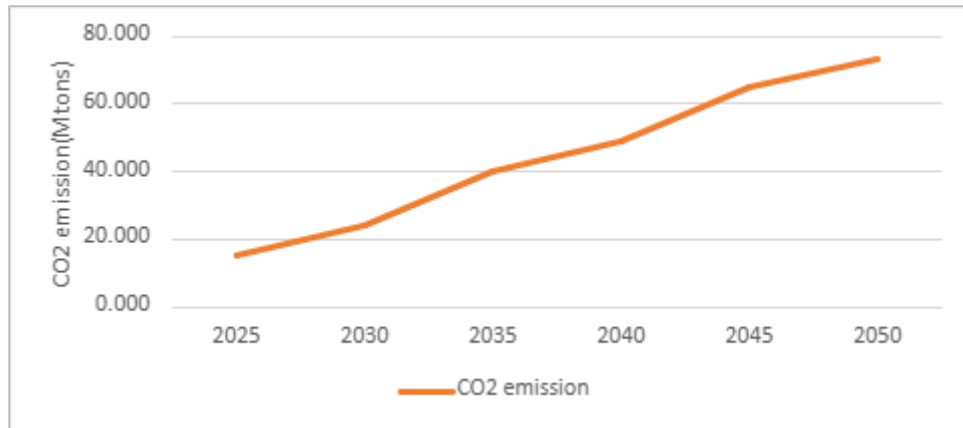


Figure 3: Overall CO2 emissions - Baseline

Without the interface of any energy policy, CO2 emissions will keep growing almost linearly. In 2050 over 73 Mtons of annual CO2 emission is predicted.

0.3. Carbon Intensity of Electricity Generation

Baseline	2025	2030	2035	2040	2045	2050
CO2 (kg)	15017180	23887850	39815500	48713830	64705520	73384780
Electricity (MWh)	122808.333	157000.000	193898.333	230000.000	264983.333	300527.778
Carbon intensity (kg/MWh)	122.281	152.152	205.342	211.799	244.187	244.186

Table 3: Carbon intensity of Electricity generation - Baseline

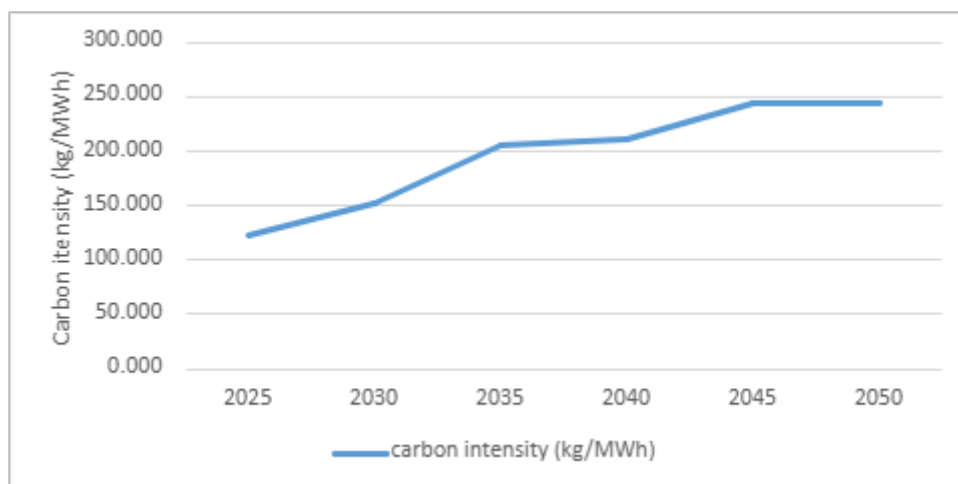


Figure 4: Carbon intensity of Electricity generation - Baseline

Carbon intensity also grows with the year, by 2050 around 244 kg/MWh of CO₂ is anticipated yearly.

0.4. New Electrical Capacities

Baseline (GW)	2025	2030	2035	2040	2045	2050
ELE_NEW_CCGT				22.39	6.61	3.89
ELE_NEW_WIND_ON	12.79	6.88	1.52		8.26	12.48
Total	12.79	6.88	1.52	22.39	14.87	16.37

Table 4: New Electrical Capacities - Baseline

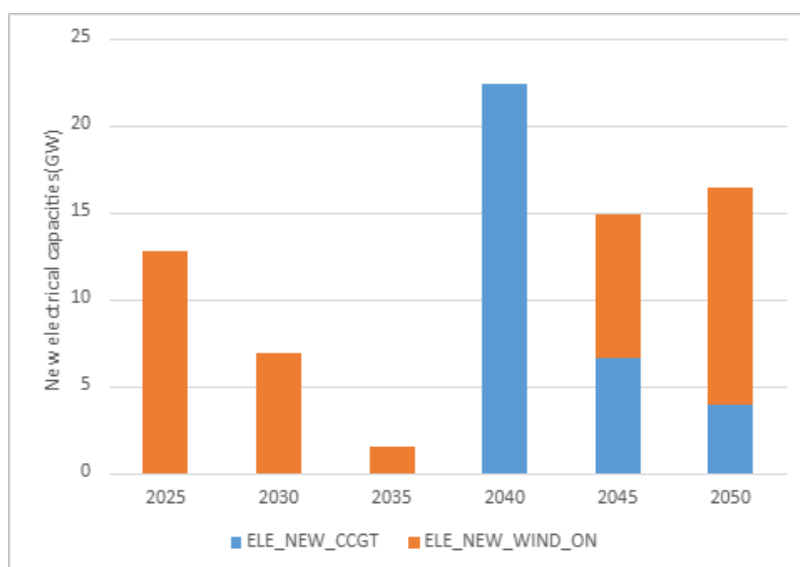


Figure 5: New Electrical Capacities - Baseline

New onshore wind plant will be installed in 2025, 2030, 2035, 2045 and 2050, new CCGT will be installed in 2040, 2045 and 2050.

0.5. Full Load Operation Hours of CCGT

Baseline	2025	2030	2035	2040	2045	2050
Annual activity (PJ)	152.9	243.22	405.39	575.7	764.7	867.27
Capacity (GW)	24.044	24.044	24.044	22.39	29.017	32.91
Full load hours	1766.44	2809.89	4683.43	7142.33	7320.42	7320.22

Table 5: Full load operation hours of CCGT - Baseline

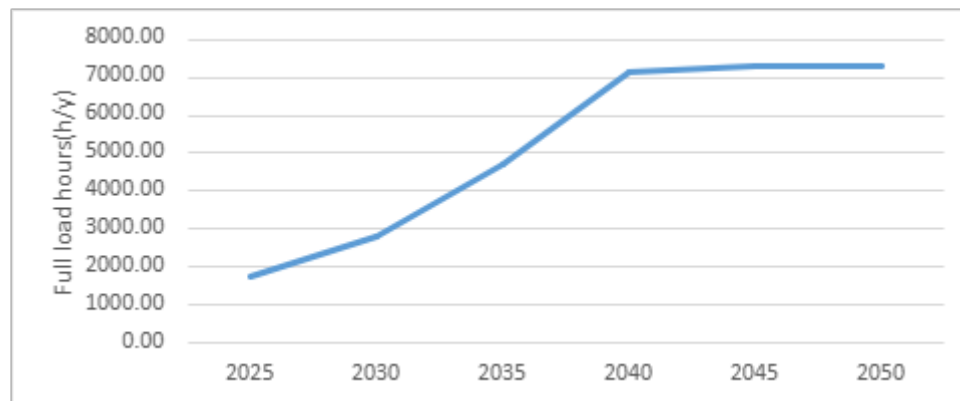


Figure 6: Full load operation hours of CCGT - Baseline

Full load hours of CCGT keep rising and arrive at its maximum level from 2045, the value is around 7320 hours/year. This indicates heavy power generation from fossil fuel.

0.6. RES Curtailment

Baseline	2025	2030	2035	2040	2045	2050
PV (TWh)	0.786691556	0	0	0	/	/
PV (%)	3.65%	0%	0%	0%	/	/
Wind OFF (TWh)	0.3016	0.0059	0.0058	0.0029	0	0
Wind OFF (%)	2.41%	0.06%	0.09%	0.09%	0%	0%
Wind ON (TWh)	1.073	1.099	1.019	0.950	0.852	0.958
Wind ON (%)	2.22%	1.75%	1.62%	1.60%	1.65%	1.64%

Table 6: Renewable Energy Sources Curtailment- Baseline

0.7. RES Share in Final Electricity Demand

RES Share	2025	2030	2035	2040	2045	2050
Baseline	65.42%	55.65%	40.89%	29.14%	19.15%	19.15%

Table 7: Renewable Energy Sources share in Final Electricity Demand - Baseline

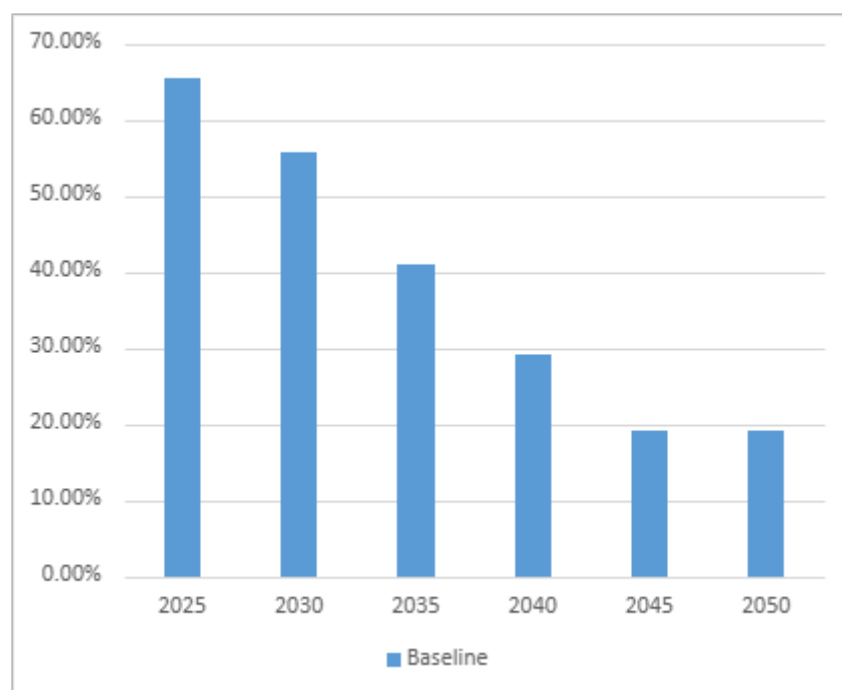


Figure 7: Renewable Energy Sources share in Final Electricity Demand - Baseline

In the baseline scenario, RES share declines from 60% to 16% during the year 2025-2050.

0.8. Electricity Generation Cost

Baseline	Item	2025	2030	2035	2040	2045	2050
CCGT	Cost (MEuro)	786.680	967.320	1291.66	3285.32	4293.89	4869.85
	Electricity (TWh)	42.472	67.561	112.608	159.917	212.417	240.908
	Electricity cost (Euro/MWh)	18.522	14.318	11.470	20.544	20.214	20.215
PV	Cost (MEuro)	205.360	154.022	102.680	51.340	0.000	0.000
	Electricity (TWh)	20.794	16.300	10.967	5.483	0.000	0.000
	Electricity cost (Euro/MWh)	9.876	9.449	9.363	9.363	0.000	0.000
Wind off	Cost (MEuro)	29.270	21.950	14.630	7.310	0.000	0.000
	Electricity (TWh)	12.232	9.394	6.261	3.131	0.000	0.000
	Electricity cost (Euro/MWh)	2.393	2.336	2.337	2.335	0.000	0.000
Wind on	Cost (MEuro)	2064.68	3131.19	3356.40	3342.33	2616.82	3496.06

	Electricity (TWh)	47.308	61.684	62.053	58.417	50.736	57.542
	Electricity cost (Euro/MWh)	43.643	50.762	54.089	57.215	51.577	60.757

Table 8: Electricity Generation Cost - Baseline

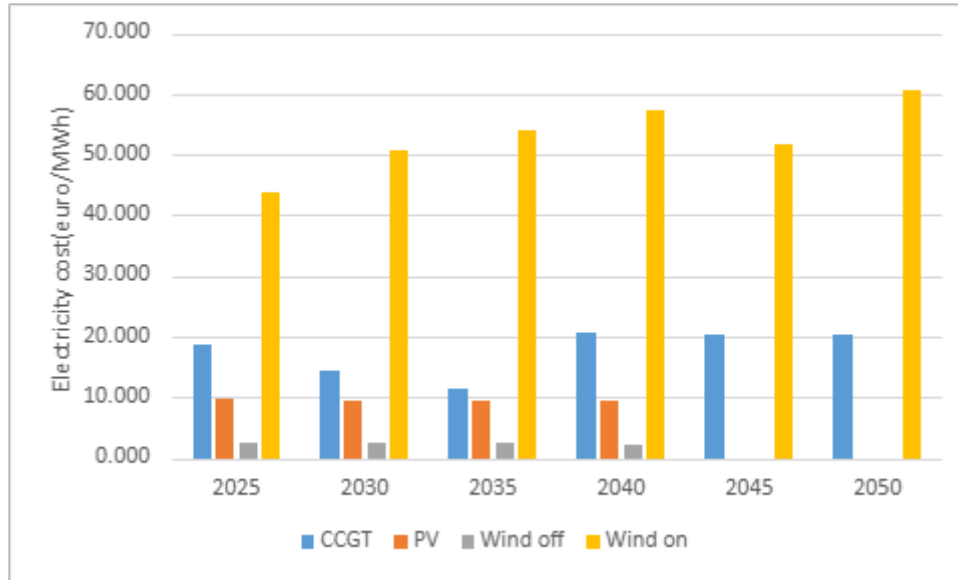


Figure 8: Electricity Generation Cost - Baseline

The ranking of electricity generation cost of technologies from low to high is offshore wind, PV, CCGT and onshore wind. The onshore wind power ranks at the highest cost because of the new capacity installation.

1. Policy 1: Carbon Tax

The first scenario under involves implementing a CO₂ tax. The implementation of a carbon tax establishes a direct financial value for each ton of carbon dioxide emissions, motivating businesses and consumers to decrease their carbon impact choosing cleaner energy alternatives. The generated revenue stream can support renewable energy development while reducing household tax liabilities through rebate programs thus creating a transparent economic solution for climate change mitigation.

1.1. Electricity Generation Mix [TWh]

Table 9 and Table 10 shows the results of the first study, where VAR_Cap values are maintained in GW, while VAR_Fout results were converted in TWh.

Process	Attribute	2025	2030	2035	2040	2045	2050
ELE_EX_GAS_CCGT	VAR_Cap (GW)	24.044	24.044	24.044			
ELE_EX_PV	VAR_Cap (GW)	20.54	15.402	10.268	5.134		

ELE_EX_WIND_OFF	VAR_Cap (GW)	2.93	2.195	1.464	0.732		
ELE_EX_WIND_ON	VAR_Cap (GW)	5.63	4.22	2.814	1.41		
ELE_NEW_WIND_ON	VAR_Cap (GW)	17.81	42.01	67.22	90.73	110	124.74

Table 9: Electricity capacity - Carbon tax

Process	Attribute	2025	2030	2035	2040	2045	2050
ELC_FIN_DEM	VAR_Fout (TWh)	122.81	157.00	193.90	230.00	264.99	300.53
ELE_EX_GAS_CCGT	VAR_Fout (TWh)	30.31	23.02	5.83			
ELE_EX_PV	VAR_Fout (TWh)	20.10	14.44	8.80	4.20		
ELE_EX_WIND_OFF	VAR_Fout (TWh)	11.77	6.34	3.21	1.29		
ELE_EX_WIND_ON	VAR_Fout (TWh)	13.93	6.60	3.18	1.26		
ELE_NEW_WIND_ON	VAR_Fout (TWh)	45.41	104.62	153.77	194.86	229.51	260.29
T&D_GRID	VAR_Fout (TWh)	31.60	24.99	24.94	28.39	35.48	40.24
T&D_GRID_RES	VAR_Fout (TWh)	91.21	132.01	168.95	201.61	229.51	260.29

Table 10: Electricity mix - Carbon tax

The CO₂ tax scenario leads to a structural change in the Dutch power sector. The 24 GW combined-cycle gas turbines (CCGT) capacity remains stable until 2050 because the tax does not make these assets economically unviable for retirement. The carbon tax then reduces CCGT operational hours to less than 6 TWh by 2035. On the other hand, the existing photovoltaic capacity decreases from 20 GW (23 TWh) to 5 GW (4 TWh) by 2040, while onshore and offshore wind decreases from 2 GW/11.7 TWh to 0.7 GW/1.3 TWh. On shore wind is the only technology showing a contrary pattern, mainly driven by the new investments which let it arrives to 124 GW installed and 260 TWh in 2050, and balancing the retirement of fossil fuels.

The system's generation mix becomes highly concentrated because of this limited investment approach, which leads to new onshore wind supplying 85% of domestic electricity output in 2050 while gas generation becomes extinct. The increasing demand creates a more difficult situation because electricity usage rises from 123 TWh in 2025 to exceed 300 TWh by 2050.

The results demonstrate that a carbon tax alone effectively moves power generation away from fossil fuels while promoting large-scale onshore wind development.

1.2. Overall GHG Emissions

Commodity	Process	Attribute	2025	2030	2035
CO2	CCGT	VAR_Fout (Mt)	10.7168407	8.14054757	2.06026617

Table 11: Overall CO2 emissions - Carbon Tax

Net emissions, which come from the CCGT plants, decrease from roughly 10.7 Mt in 2025 to 8.1 Mt in 2030. The change is mainly driven by the reduced utilization of gas plants under the tax. By 2035, as onshore wind now supplies a large share of generation and gas consumptions decrease, CO₂ emissions from CCGT fall to just over 2 Mt. Beyond that point, no remaining gas-fired generation remain in the model, and the power-sector emissions approach zero. These results show the carbon tax effectively compresses roughly 80% of emissions in 15 years.

1.3. Carbon Intensity of Electricity Generation

To calculate the carbon intensity of electricity generation [kg/MWh], the CCGT plant CO₂ emissions were divided by the total VAR_FOut output. Table 12 shows the results. It can be seen how the on-going switch from the CCGT to the onshore wind is reflected in a lower and lower carbon intensity in the electricity mix in 2035. After this date, no information was provided by the software regarding the CCGT.

Attribute	2025	2030	2035
Carbon intensity (kgCO2/MWh)	87.265	51.851	10.625

Table 12: Carbon intensity of electrical generation - Carbon Tax

1.4. New Electrical Capacities

Process	Attribute	2025	2030	2035	2040	2045	2050
New Wind On	VAR_Ncap (GW)	17.818	24.201	25.198	23.5127	32.9475	43.0865

Table 13: New electrical capacities - Carbon Tax

As analysed in the electricity generation mix study, the only sector that improves its capacity it is the onshore wind. In Table 13 it is shown the marginal increment per milestone year, with a total capacity installed of 124.75 GW in 2050.

1.5. Full Load Operation Hours of CCGT

To find the full operation hours a computation is needed. Particularly, after extracting for the CCGT process the VAR_Cap (capacity, in GW) and VAR_FOut (generation, converted to TWh) attributes, by dividing them it is possible to find the full-load hours per year. A conversion from GW to TW was needed for the VAR_Cap results.

The full-load hours are shown in Table 14.

Process	Attribute	2023	2025	2030	2035
ELE_EX_GAS_CCGT	VAR_Cap (TW)	0.024044	0.024044	0.024044	0.024044
ELE_EX_GAS_CCGT	VAR_Fout (TWh)	52.85	30.31	23.02	5.83
ELE_EX_GAS_CCGT	full-load hours	2198.053	1260.6055	957.41141	242.47213

Table 14: Full load operation hours of CCGT - Carbon tax

1.6. RES Curtailment

RES Curtailment (TWh)	2025	2030	2035	2040	2045	2050
PVs	1.48	1.74	1.99	1.19		
PVs (%)	6.85%	10.77%	18.46%	22.12%		
Offshore Wind	0.77	3.06	3.06	1.84		
Offshore Wind (%)	6.12%	32.60%	48.85%	58.87%		
Onshore Wind	2.24	10.27	27.09	46.00	59.57	67.52
Onshore Wind (%)	3.63%	8.45%	14.72%	19.00%	20.61%	20.60%

Table 15: Renewable Energy Sources Curtailment - Carbon Tax

Under a stand-alone CO₂ tax, on shore wind is during the years the technology with the highest energy curtailed. Its trend rises progressively from 2025 to 2050, while Table 15 shows how PVs and offshore wind reach lower values, even roughly zero.

1.7. RES Share in Final Electricity Demand

The VAR_FOut values presented are already converted in TWh.

Commodity	Process	2025	2030	2035	2040	2045	2050
ELC_FIN	ELC_FIN_DEM	122.81	157.00	193.90	230.00	264.98	300.53
ELC_GRID	T&D_GRID	31.60	24.99	24.94	28.39	35.48	40.24
ELC_GRID_RES	T&D_GRID_RES	91.21	132.01	168.95	201.61	229.51	260.29

Table 16: Renewable Energy Sources share in final electricity demand - Carbon Tax

The standalone carbon-tax scenario shows how the renewables expand their role in meeting final electricity demand. In 2025, renewable generation delivered to the grid 91.2 TWh satisfying roughly 74 % of total demand (122.8 TWh). By 2030, that share rises to about 84 % as wind accelerate (122.81 TWh of RES vs. 157.0 TWh total). Renewables peak near 87 % of demand in the mid 2030s, delivering 168.9 TWh against 193.9 TWh of consumption, to then hold between 86%–88 % through 2050 (260.3 TWh of RES vs. 300.5 TWh total). This trend

highlights how a carbon tax strongly drives both the build-out and use of renewable generation, while leaving a small 12–14 % slice of power to conventional, non-renewable sources.

1.8. Electricity Generation Cost

Baseline	Item	2025	2030	2035	2040	2045	2050
CCGT	Cost (MEuro)	699.11	646.65	522.83			
	Electricity (TWh)	30.31	23.02	5.83			
	Electricity cost (Euro/MWh)	23.07	28.09	89.68			
PV	Cost (MEuro)	205.36	154.02	102.68	51.34		
	Electricity (TWh)	20.10	14.44	8.80	4.20		
	Electricity cost (Euro/MWh)	10.22	10.67	11.67	12.22		
Wind off	Cost (MEuro)	29.27	21.95	14.63	7.31		
	Electricity (TWh)	11.77	6.34	3.21	1.29		
	Electricity cost (Euro/MWh)	2.49	3.46	4.56	5.67		
Wind on	Cost (MEuro)	2798	6597	10554	14246	16621	19586
	Electricity (TWh)	59.34	111.22	156.95	196.1	229.5	260.29
	Electricity cost (Euro/MWh)	47.15	59.32	67.24	72.64	72.42	75.25

Table 17: Electricity generation cost - Carbon Tax

Table 17 underlines how the most cost-effective technology is the offshore wind, followed by PVs, CCGT and finally on shore wind, which costs however consider the investments to build the structure.

1.9. Subsidies and Taxes Values

The imposed taxes' values can be found in the Scen_TAX_CO2 Excel file, and here shown in Table 18.

Attribute	Year	AllRegions	Cset_CN
\I:		MEur/kton	
COM_TAXNET	2023	0.07	CO2
COM_TAXNET	2030	0.1	CO2
COM_TAXNET	2040	0.2	CO2
COM_TAXNET	2050	0.3	CO2
COM_TAXNET	0	5	CO2

Table 18: Taxes values - Carbon Tax

The Table 18 shows a progressively rising CO₂ tax over the modeling horizon. In 2023, the tax is set at 0.07 MEUR per kton of CO₂ (i.e. €70 / tCO₂), increasing to 0.10 MEUR/kton (€100 / tCO₂) by 2030, 0.20 MEUR/kton (€200 / tCO₂) by 2040, and reaching 0.30 MEUR/kton (€300 / t) by 2050. This progressive escalation is intended to incentives for emissions reductions and driving gradual decarbonization of the power sector. The year 0 with a MEur/kton was set for VEDA interpolation purposes.

2. Policy 2: Cap on CO₂ Emissions

The second policy focuses on fixing an absolute ceiling on the total amount of carbon that can be released into the atmosphere. Under a cap-and-trade framework, government sets a declining annual limit on economy-wide CO₂ emissions and issues tradable permits—each representing the right to emit one metric ton of carbon dioxide. In this study, the cap is set at 16 Mtons of CO₂ from 2025 to 2050. Because the number of permits is capped, the policy guarantees that aggregate emissions stay within the scientific pathway required to meet climate targets.

By turning the right to pollute into a scarce commodity, a cap harnesses market forces to discover the lowest-cost abatement opportunities. Firms that can cut emissions more cheaply do so and sell their surplus permits; those facing higher reduction costs buy permits on the secondary market, ensuring that every ton of CO₂ is reduced where it is most economical. Regular auctioning of allowances can create a predictable price signal that rewards energy efficiency, accelerates the shift toward renewables, and spurs innovation in low-carbon technologies.

Auction revenues offer additional policy leverage: funds can be earmarked for grid modernization, green infrastructure, and dividend or rebate programs that cushion households—particularly low- and middle-income ones—against any rise in energy prices. In this way, a cap on emissions provides a transparent, enforceable, and market-based pathway to climate stability while supporting equitable economic transitions.

2.1. Electricity Generation Mix [TWh]

Policy 2 (TWh)	2025	2030	2035	2040	2045	2050
CCGT	42.47	45.25	46.49	52.525	52.525	52.525
Wind ON	47.25	84.62	129.48	162.47	201.24	234.60
Wind OFF	12.23	8.95	4.8	2.19	0	0
PV	20.86	16	10.2	5.00	0	0
Import	0	2.17	2.91	7.82	11.22	13.4
Total Electricity demand	122.81	157	193.90	230	264.98	300.53

Table 19: Electricity Generation Mix (TWh) - Cap on CO₂ Emissions

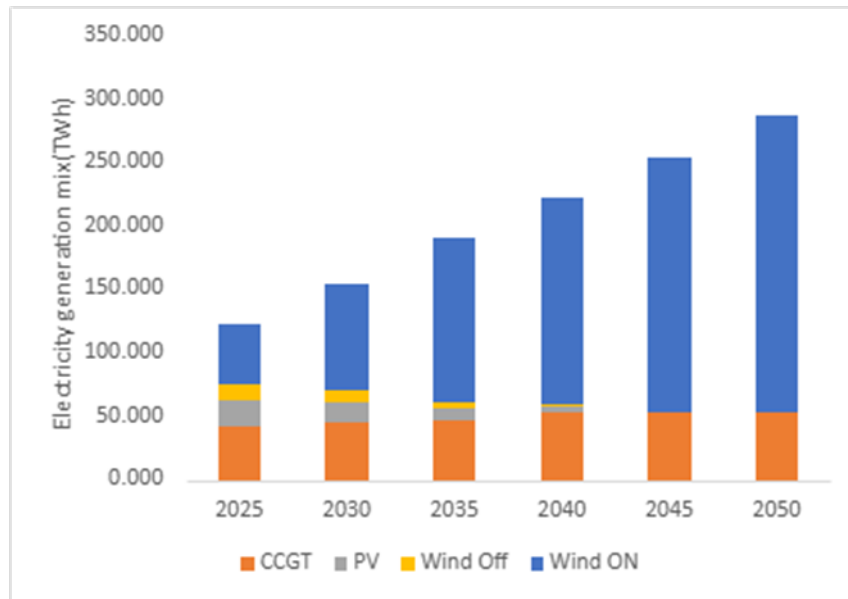


Figure 9: Electricity Generation Mix (TWh) - Cap on CO2 Emissions

The electricity generation mix becomes more and more sustainable throughout the years. Indeed, the limit of CO2 imposed the electricity production from CCGT to be below or equal to 52.525 TWh in 5 years in order to satisfy the CO2 limit imposed. As such, much onshore wind power is installed to satisfy electrical demand. It can also be noted that the imported electricity increases throughout the years, but remains at a relatively low level.

2.2. Overall GHG Emissions

Policy 2 (Mtons)	2025	2030	2035	2040	2045	2050
CO2 emissions	15.017	16	16	16	16	16

Table 20: Overall CO2 Emissions - Cap on CO2 Emissions

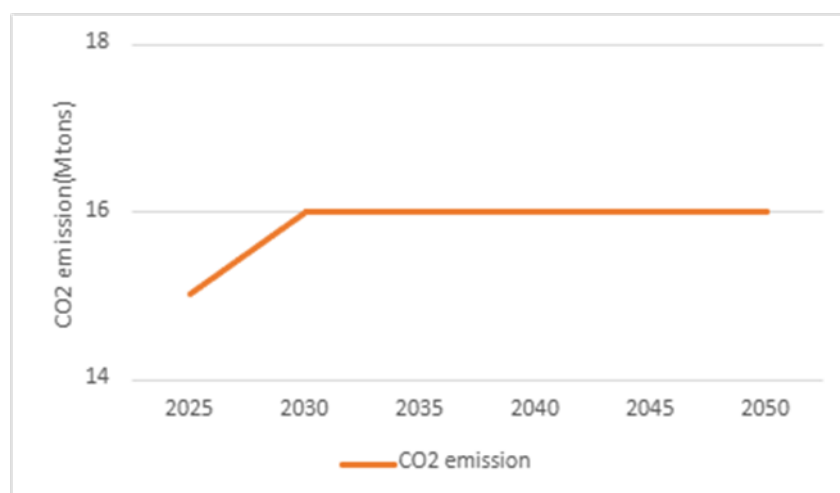


Figure 10: Overall CO2 Emissions - Cap on CO2 Emissions

From these results, the limit of CO2 emissions is set at 16 Mt every year. We can see that the emissions are below this threshold limit in 2025. This is because there are many already

installed renewable energy sources that are producing substantial amounts of electricity. The next years, the renewables need to be replaced, and the use of CCGT is therefore maximized while staying within the emissions limit.

2.3. Carbon Intensity of Electricity Generation

Policy 2	2025	2030	2035	2040	2045	2050
Carbon intensity of electricity generation [kg/MWh]	122.28	101.91	82.52	69.56	60.38	53.24

Table 21: Carbon intensity of electricity generation - Cap on CO2 Emissions

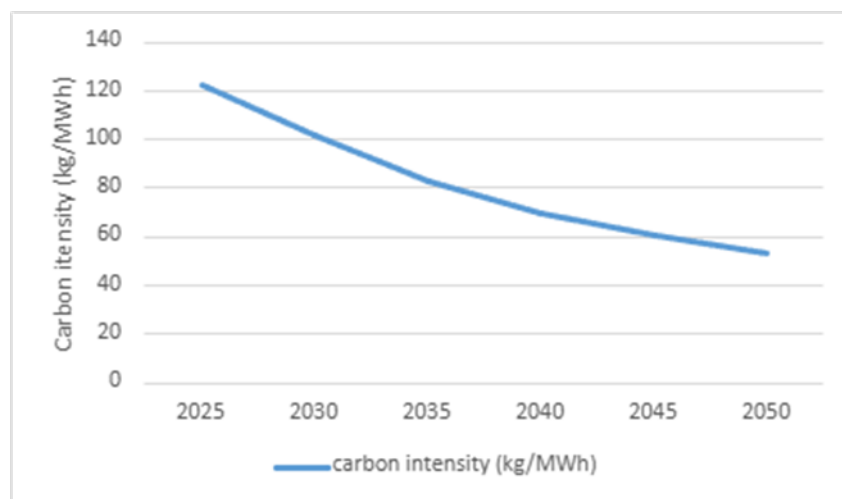


Figure 11: Carbon intensity of electricity generation - Cap on CO2 Emissions

The carbon intensity of the electricity generation becomes smaller as the years go by. Indeed, the final energy demand grows year after year, while the total yearly emissions don't increase, due to the CO2 emissions limit policy.

2.4. New Electrical Capacities

Policy 2 (GW)	2025	2030	2035	2040	2045	2050
ELE_NEW_CCGT			1.33	10.44	0.67	0.61
ELE_NEW_WIND_ON	12.79	16.24	20.29	15.9	30.18	35.07

Table 22: New Electrical Capacities - Cap on CO2 Emissions

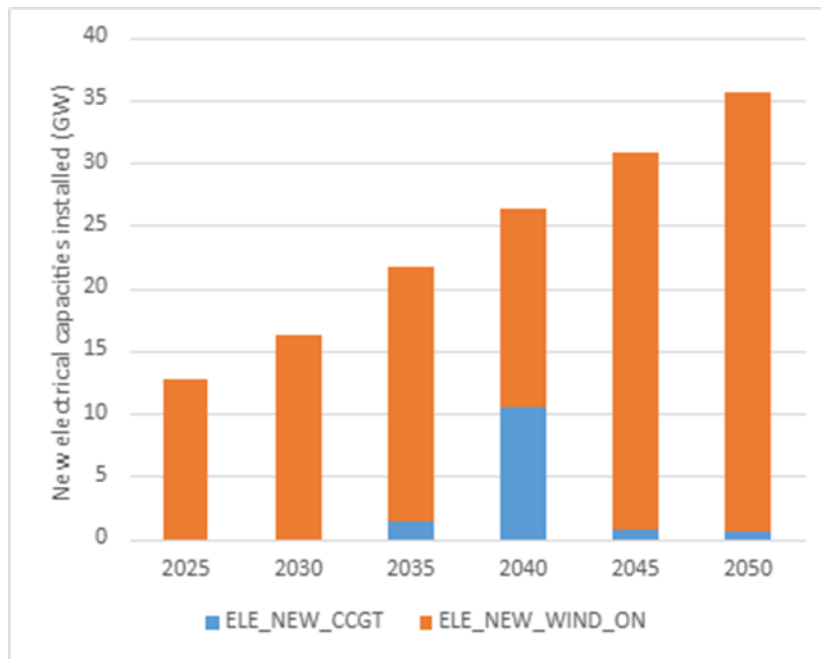


Figure 12: New Electrical Capacities - Cap on CO2 Emissions

It is important to note that the values displayed are the capacities installed for each year.

New CCGT powerplants are installed from 2035, in order to compensate for the reduction in energy generation of the previously installed CCGT as they approach the end of their lifetime. This is why most of the CCGT is installed in 2040, while there is nearly no installation in the other years.

New onshore Wind powerplants are thus installed, already from 2025, in order to generate the needed electricity, in order to satisfy demand while ensuring no new sources of CO2 emissions.

2.5. Full Load Operation Hours of CCGT

Baseline	2025	2030	2035	2040	2045	2050
Annual activity (TWh)	42.47	45.25	46.49	52.525	52.525	52.525
Capacity (TW)	0.024044	0.024044	0.025374	0.01178	0.01245	0.013062
Full load hours	1766.4	1882	1832	4459	4219	4021

Table 23: Full load operation hours of CCGT - Cap on CO2 Emissions

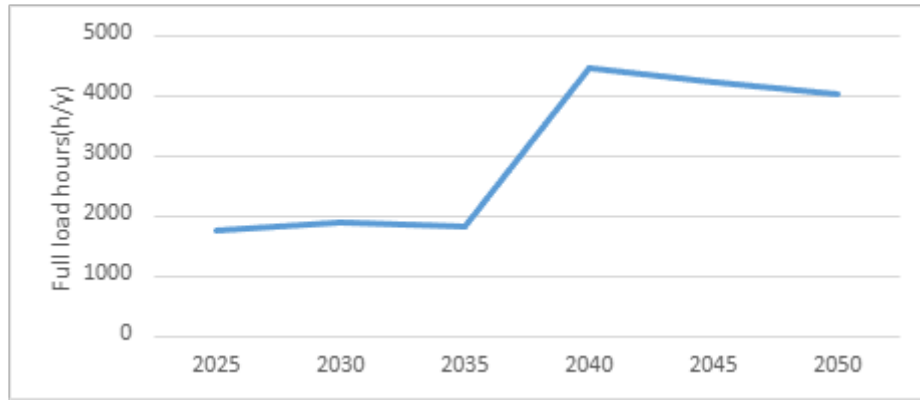


Figure 13: Full load operation hours of CCGT - Cap on CO2 Emissions

From these results, it is possible to observe the cost optimization made when installing the capacities for the CCGT in 2040. As a matter of fact, before this date, the CCGT were used until the CO2 emissions reached the allowed limit. But as the CCGT need to be replaced, only necessary capacity is installed, and the plants run much more each year to produce necessary electricity.

2.6. RES Curtailment

In order to analyse the RES curtailment that occurs, a focus will be done studying each renewable energy source: Solar, onshore Wind and offshore Wind. For each of these sources, the total potential of electricity produced will be computed and compared to the actual electricity produced.

Policy 2	2025	2030	2035	2040	2045	2050
PV (TWh)	0.723080444	0.18848	0.571423111	0.395433778	/	/
PV (%)	3.35%	1.16%	5.30%	7.33%	/	/
Wind OFF (TWh)	0.3019	0.4531	1.4669	0.9473	0.0000	0.0000
Wind OFF (%)	2.41%	4.82%	23.41%	30.23%	0%	0%
Wind ON total (TWh)	1.134	2.772	7.541	12.634	23.689	32.034
Wind ON (%)	2.34%	3.17%	5.50%	7.22%	10.53%	12.01%

Table 24: Renewable Energy Sources Curtailment - Cap on CO2 Emissions

Interestingly, the curtailment rate raises throughout the years, and even reaches 12% in 2050 for onshore wind electricity production. This leaves a margin for better energy management, as the curtailed energy could be used by flexible loads, or could be stored for later use in storage solutions.

2.7. RES Share in Final Electricity Demand

The share of renewable energy sources in final electricity demand has been done comparing the final energy demand to the addition of the electricity produced by each renewable energy source. The final electricity demand also incorporates the imported energy (even if it should now be considered, it is, as it is only a small portion of the total energy supply).

Policy 2	2025	2030	2035	2040	2045	2050
Final electricity demand (in PJ)	442.11	565.2	698.03	828	953.94	1081.9
Total RES electricity generation (in PJ)	289.21	394.46	520.2	610.76	724.47	844.57
Share of RES in final electricity demand	65.42%	69.79%	74.52%	73.76%	75.95%	78.06%

Table 25: Share of Renewable Energy Source in final electricity demand - Cap on CO2 Emissions

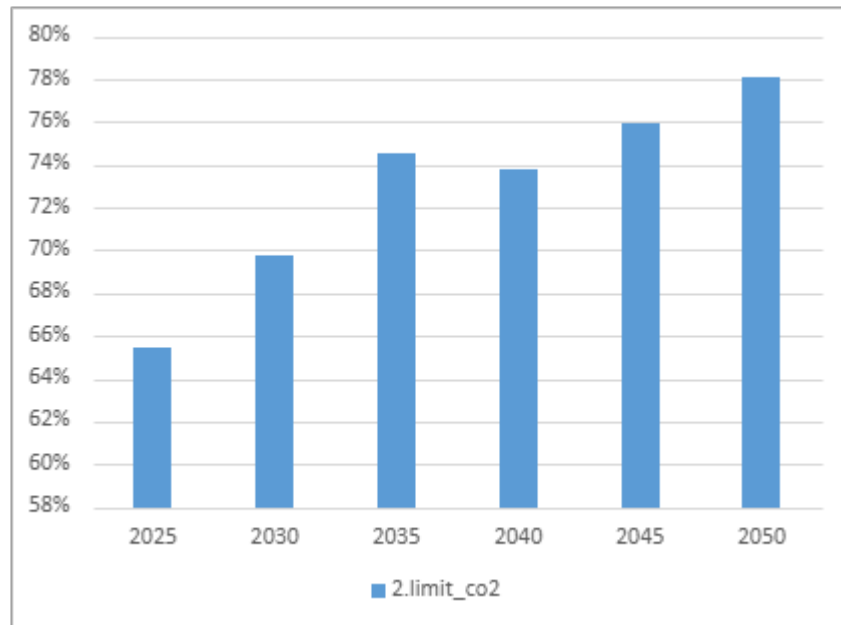


Figure 14: Share of Renewable Energy Source in final electricity demand - Cap on CO2 Emissions

As expected, imposing a limit on the CO2 emissions increases the share of RES in final electricity demand, as alternatives to conventional powerplants need to be installed in order to ensure energy demand is satisfied. It is also interesting to note a slight decrease in the share of renewables from 2035 to 2040, as most previously installed RES are running out of their operational lifetime, and get partially replaced by CCGT plants providing around 25% of the final energy demand in 2040. This share however decreases after the years, as the CCGT plants cannot produce more electricity in order to respect the CO2 emissions limit.

It can thus be observed that putting a limit on CO2 emissions greatly enhances the adoption and development of renewable energy sources, as 78% of electricity is provided by RES in 2050.

2.8. Electricity Generation Cost

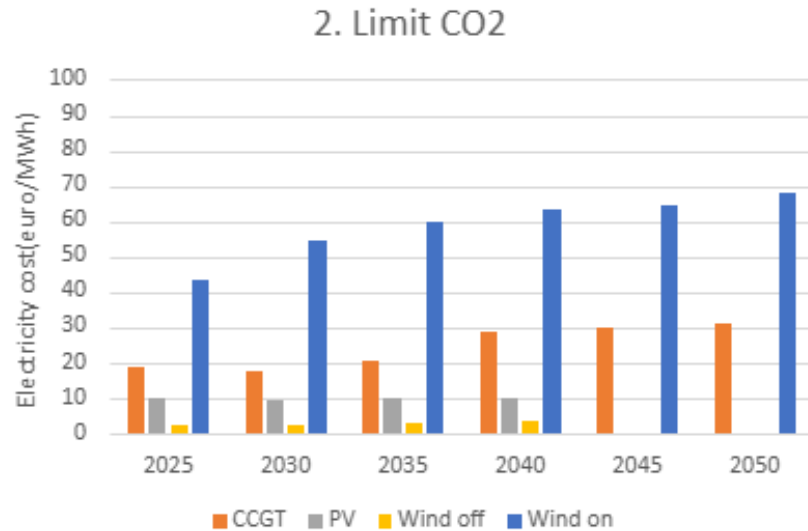


Figure 15: Electricity Generation Cost - Cap on CO2 Emissions

Baseline	Item	2025	2030	2035	2040	2045	2050
CCGT	Cost (MEuro)	786.68	806.69	942.90	1500.57	1564.48	1622.63
	Electricity (TWh)	42.47	45.25	46.49	52.53	52.53	52.53
	Electricity cost (Euro/MWh)	18.52	17.83	20.28	28.57	29.79	30.89
PV	Cost (MEuro)	205.36	154.02	102.68	51.34	0.00	0.00
	Electricity (TWh)	20.86	16.00	10.21	5.00	0.00	0.00
	Electricity cost (Euro/MWh)	9.85	9.63	10.05	10.27	0.00	0.00
Wind off	Cost (MEuro)	29.27	21.95	14.63	7.31	0.00	0.00
	Electricity (TWh)	12.23	8.95	4.80	2.19	0.00	0.00
	Electricity cost (Euro/MWh)	2.39	2.45	3.05	3.34	0.00	0.00
Wind on	Cost (MEuro)	2064.68	4601.04	7774.02	10257.37	12974.14	15930.62
	Electricity (TWh)	47.25	84.62	129.48	162.47	201.24	234.60
	Electricity cost (Euro/MWh)	43.70	54.37	60.04	63.13	64.47	67.90

Table 26: Electricity Generation Cost - Cap on CO2 Emissions

The results obtained show that onshore Wind and CCGT energy have the highest electricity cost, while PV and offshore Wind have the lowest electricity cost. This can partially be explained by the investment capital needed to install the additional amounts of onshore wind and CCGT technologies.

Moreover, it is interesting to note that the electricity cost rises throughout the years.

3. Policy 3: Investment Mandate for Specific Technologies

This scenario enforces a minimum build-out of offshore wind power, ensuring that at least 1 GW is installed by 2025 and 5 GW from 2030 to 2050. Unlike price-based policies, this mandate guarantees deployment by setting clear targets.

By requiring steady capacity additions, the policy provides long-term certainty for grid planning, supply chains, and investors. It supports cost reductions through economies of scale and helps accelerate the transition from fossil fuels to renewables. This direct approach ensures tangible progress in decarbonizing the power sector while also supporting industrial development in coastal regions.

3.1. Electricity Generation Mix [TWh]

Policy 3 (TWh)	2025	2030	2035	2040	2045	2050
CCGT	41.15	63.28	90.21	116.14	168.51	209.53
Wind ON	44.30	41.04	37.49	33.85	7.01	3.18
Wind OFF	16.41	34.95	53.34	71.60	86.58	85.59
PV	20.95	16.45	10.94	5.48	0.00	0.00
Import	0.00	1.28	1.91	2.92	2.89	2.23
Total Electricity demand	122.81	157	193.90	230	264.98	300.53

Table 27: Electricity Generation Mix (TWh) - Investment Mandate for Specific Technologies

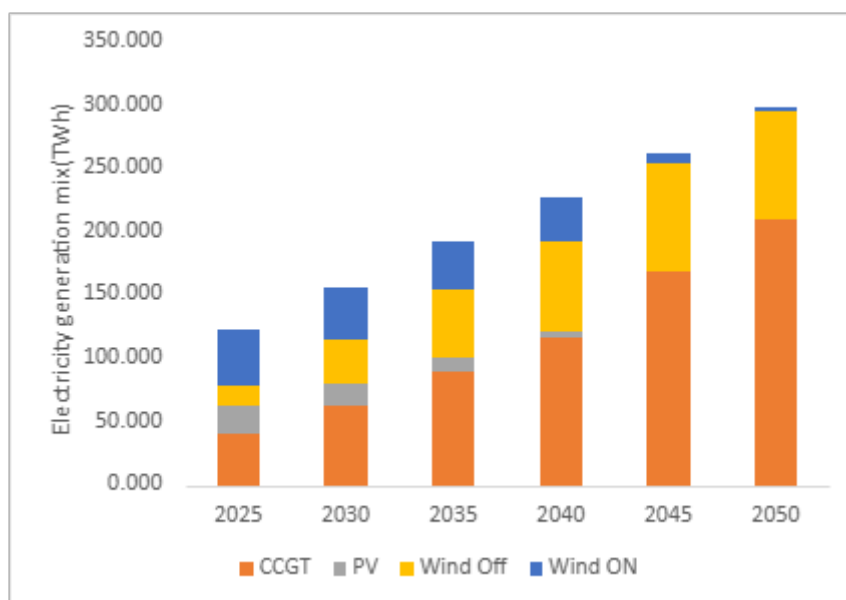


Figure 16: Electricity Generation Mix (TWh) - Investment Mandate for Specific Technologies

The electricity mix slowly gets dominated by CCGT electricity generation after the years. However, there is a raise in the electricity production from offshore Wind energy, due to the implemented policy, which imposes a minimum of 1 GW of installed offshore Wind powerplants in 2025 and 5 GW until 2050. This policy is thus interesting for raising the

implementation of offshore wind, but is not a strong enough policy to stop and replace the use of conventional powerplants such as CCGT.

It is also interesting to note that even if the investment mandates were proposed for offshore wind, there is much wind energy generated from the already installed onshore installations in the energy mix until 2040. It can also be noted that while onshore Wind installations have lower costs, offshore Wind powerplants are preferred in this model due to the investment mandate imposed.

3.2. Overall GHG Emissions

Policy 2 (Mtons)	2025	2030	2035	2040	2045	2050
CO2 emissions	14.55	22.37	31.90	35.38	51.33	63.83

Table 28: Overall CO2 Emissions (Mtons) - Investment Mandate for Specific Technologies

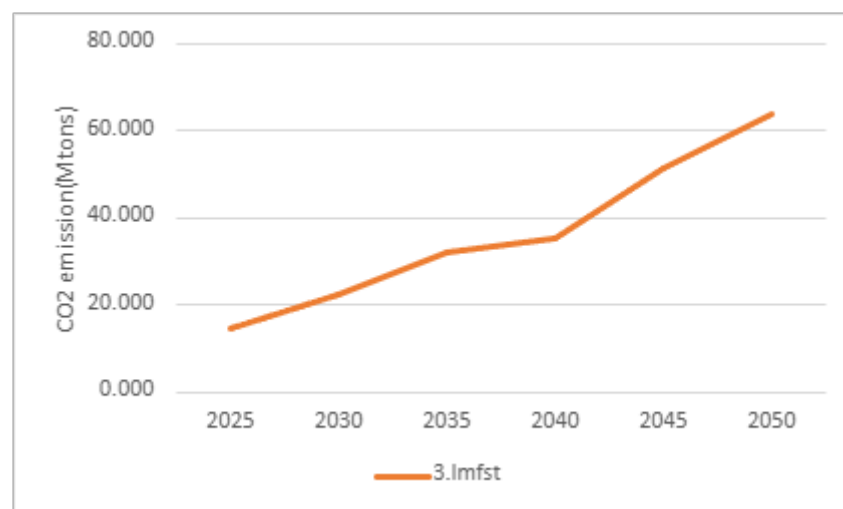


Figure 17: Overall CO2 Emissions (Mtons) - Investment Mandate for Specific Technologies

From these results, a continuous rise in CO2 emissions appear throughout the years, although a slower rise can be observed from 2035 to 2040. This can be explained by the raise in offshore wind electricity production and the steady onshore wind production in these 5 years.

3.3. Carbon Intensity of Electricity Generation

Including imported energy

Policy 3	2025	2030	2035	2040	2045	2050
Carbon intensity of electricity generation [kg/MWh]	118.48	142.51	164.50	153.81	193.71	212.38

Table 29: Carbon intensity of electricity generation - Investment Mandate for Specific Technologies

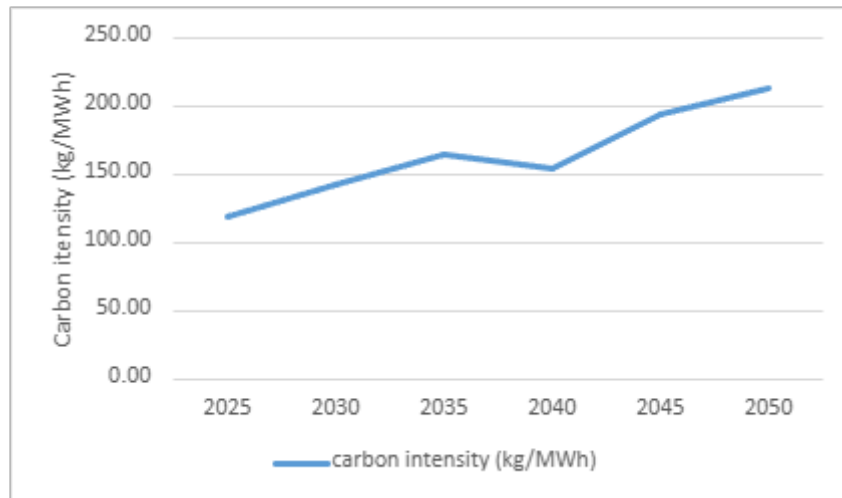


Figure 18: Carbon intensity of electricity generation - Investment Mandate for Specific Technologies

The carbon intensity of the electricity generation grows overall as the years go by, reaching 212 kg/MWh in 2050. However, a small decline in carbon intensity can be observed from 2035 to 2040. As the carbon intensity grows overall, this policy can therefore not push for the renewable energy transition without other policies coming into play, as the carbon emissions grow faster than the electricity demand.

3.4. New Electrical Capacities

Policy 3 (GW)	2025	2030	2035	2040	2045	2050
ELE_NEW_CCGT				18.004	6.12	4.99
ELE_NEW_WIND_OFF	1	5	5	5	5	5
ELE_NEW_WIND_ON	11.69					1.23

Table 30: New electrical capacities - Investment Mandate for Specific Technologies



Figure 19: New electrical capacities (GW) - Investment Mandate for Specific Technologies

New CCGT powerplants are installed from 2040, in order to compensate for the reduction in energy generation of the previously installed CCGT as they approach the end of their lifetime. Even though new onshore and offshore wind installations have been installed from 2025, the capacity installed isn't enough to supply the whole demand, therefore needing cheaper CCGT powerplants. It is however interesting to observe that in 2050, more onshore wind power is installed even though no additional capacity has been installed since 2025, most probably because these plants run out of lifetime. This therefore shows a growing trend in using renewables, even though this remains low in comparison to the whole energy demand.

3.5. Full Load Operation Hours of CCGT

Policy 3	2025	2030	2035	2040	2045	2050
Annual activity (TWh)	41.15	63.28	90.21	116.14	168.51	209.53
Capacity (GW)	0.02404	0.02404	0.02404	0.01800	0.02412	0.02912
Full load hours	1711	2632	3752	6451	6986	7195

Table 31: Full load operation hours - Investment Mandate for Specific Technologies

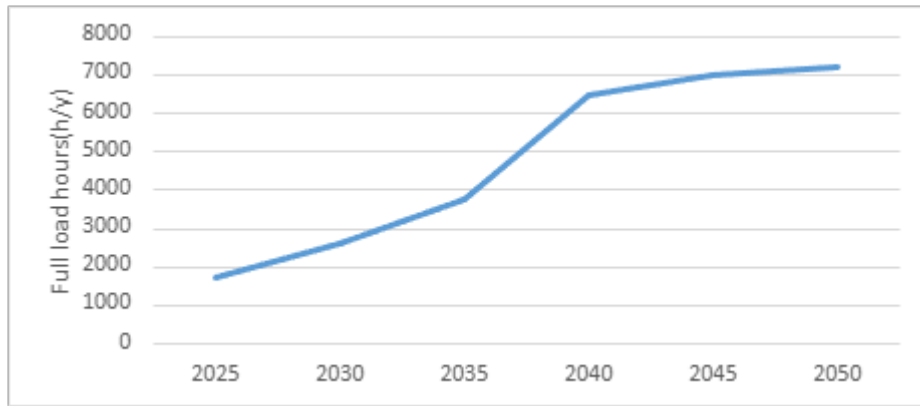


Figure 20: Full load operation hours - Investment Mandate for Specific Technologies

CCGT plants are increasing their equivalent full load hours throughout the years. These plants are thus installed and utilized to their near-maximum (at nearly 7200 full load equivalent hours) in order to ensure electrical demand is met and to reduce investment costs.

3.6. RES Curtailment

Policy 3	2025	2030	2035	2040	2045	2050
PV (TWh)	0.62836	0	0	0	/	/
PV (%)	2.91%	0%	0%	0%	/	/
Wind OFF (TWh)	0.4207	0.2075	0.1458	0.2091	0.2547	0.2591
Wind OFF (%)	2.50%	0.59%	0.27%	0.29%	0.29%	0.30%
Wind ON total (TWh)	1.193	0.775	0.617	0.548	0.116	0.049
Wind ON (%)	2.62%	1.85%	1.62%	1.59%	1.63%	1.52%

Table 32: Renewable Energy Sources Curtailment - Investment Mandate for Specific Technologies

From these results, it is possible to observe that the curtailment rates in this policy are very low, meaning that most of the renewable electricity potential is utilized to its maximum. This is very good from an environmental perspective.

3.7. RES Share in Final Electricity Demand

The share of renewable energy sources in final electricity demand has been done comparing the final energy demand to the addition of the electricity produced by each renewable energy source. The final electricity demand also incorporates the imported energy (but it is considered, as it is only a small portion of the total energy supply).

Policy 3	2025	2030	2035	2040	2045	2050
Final electricity demand (in PJ)	442.11	565.2	698.034	828	953.94	1081.9
Total RES electricity generation (in PJ)	293.97	332.78	366.38	399.39	336.92	319.58
Share of RES in final electricity demand	66.49%	58.88%	52.49%	48.24%	35.32%	29.54%

Table 33: Share of Renewable Energy Sources in final electricity demand - Investment Mandate for Specific Technologies

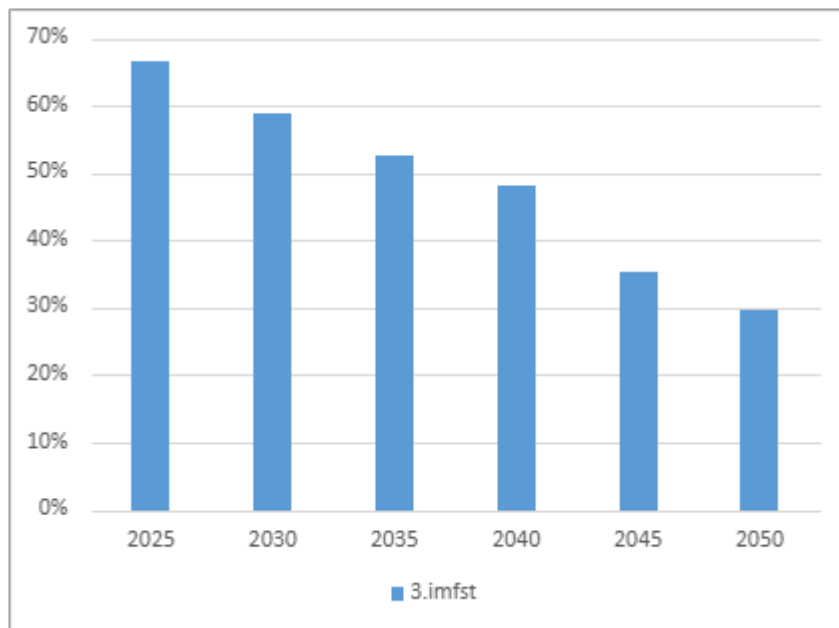


Figure 21: Share of Renewable Energy Sources in final electricity demand - Investment Mandate for Specific Technologies

As it could be expected from the energy mix, the RES share in the final electricity consumptions reduces throughout the years, as more CCGT plants are running and producing non-renewable energy. Even with a mandate for renewable energy installations, the renewable share decreases, reaching 30% of electricity generation in 2050.

However, even given this downwards trend, around one third of the consumed electricity is still provided by renewable energy sources. With proper additional policies, Investment Mandate for Specific Technologies is a good way to enhance the integration of renewable in electricity generation.

3.8. Electricity Generation Cost

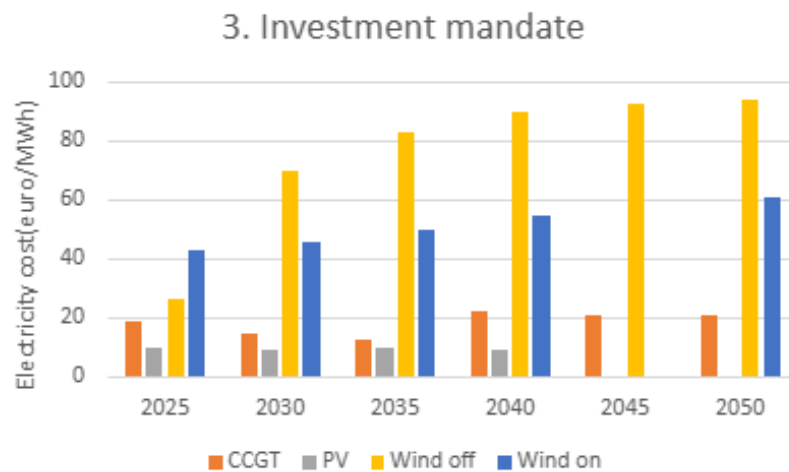


Figure 22: Electricity generation cost - Investment Mandate for Specific Technologies

Policy 3	Item	2025	2030	2035	2040	2045	2050
CCGT	Cost (MEuro)	777.17	936.50	1130.40	2551.41	3512.05	4283.21
	Electricity (TWh)	41.15	63.28	90.21	116.14	168.51	209.53
	Electricity cost (Euro/MWh)	18.89	14.80	12.53	21.97	20.84	20.44
PV	Cost (MEuro)	205.36	154.02	102.68	51.34	0.00	0.00
	Electricity (TWh)	20.95	16.45	10.94	5.48	0.00	0.00
	Electricity cost (Euro/MWh)	9.80	9.36	9.38	9.36	0.00	0.00
Wind off	Cost (MEuro)	431.30	2434.12	4436.95	6439.77	8040.58	8040.58
	Electricity (TWh)	16.41	34.95	53.34	71.60	86.58	85.59
	Electricity cost (Euro/MWh)	26.29	69.65	83.19	89.94	92.87	93.94
Wind on	Cost (MEuro)	1892.23	1878.16	1864.09	1850.03	0.00	193.50
	Electricity (TWh)	44.30	41.04	37.49	33.85	7.01	3.18
	Electricity cost (Euro/MWh)	42.72	45.77	49.72	54.65	0.00	60.79

Table 34: Electricity generation cost - Investment Mandate for Specific Technologies

The results show that the technologies with highest electricity cost are onshore and offshore wind, while the technologies with the cheapest electricity cost are PV and CCGT. It is also interesting to note that the cost of each technology rises throughout the years.

4. Policy 4: Capital Subsidies

4.1. Electricity Generation Mix [TWh]

Subsidy(Twh)	2025	2030	2035	2040	2045	2050
CCGT	24.839	20.078	20.570	21.086	25.854	29.319
PV	18.881	14.669	9.406	4.408	0.000	0.000
Wind Off	64.753	113.431	158.408	199.411	233.658	265.001
Wind ON	13.558	7.844	4.128	1.717	0.000	0.000

Table 35: Electricity generation mix - Capital Subsidies

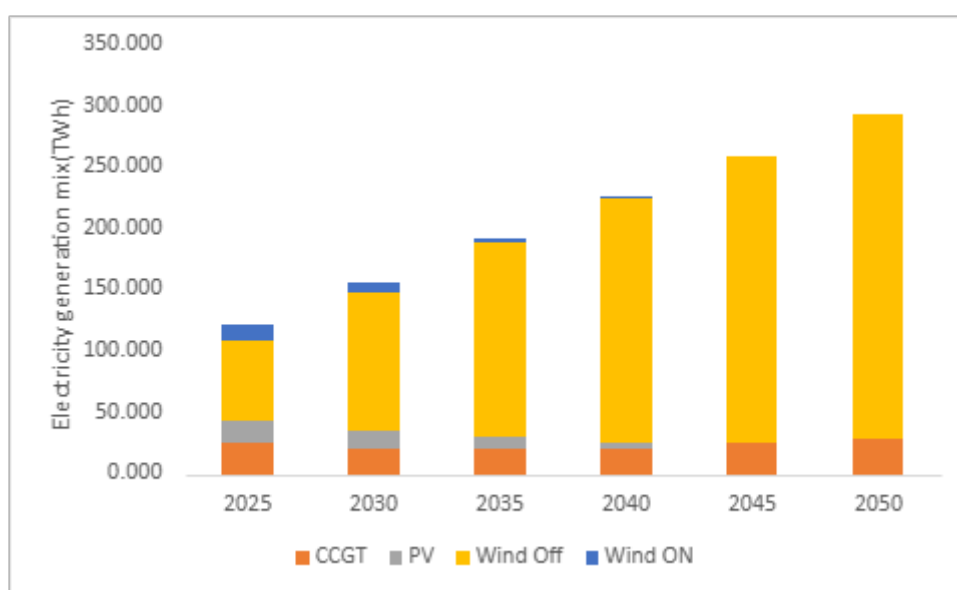


Figure 23: Electricity generation mix - Capital Subsidies

With 2000euro/kW subsidy to new offshore wind power, most of the electricity generated by 2050 would come from offshore wind, arriving 234TWh. Meanwhile, there will be no power generated from PV and onshore wind from 2045. CCGT will still run but only provide 52 TWh annual electricity as compensation for the energy system.

4.2. Overall GHG Emissions

Subsidies (Mtons)	2025	2030	2035	2040	2045	2050
CO2 emission	8.783	7.100	7.273	6.424	7.876	8.932

Table 36: Overall CO2 Emissions - Capital Subsidies

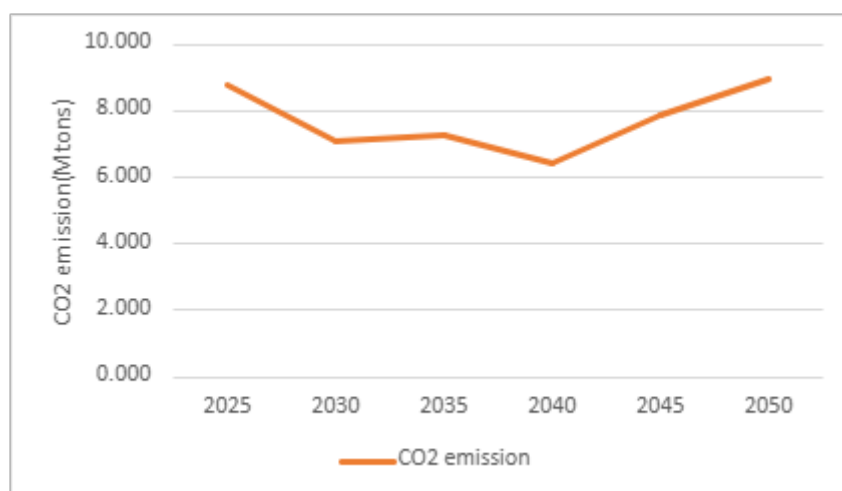


Figure 24: Overall CO2 Emissions - Capital Subsidies

With the offshore wind power subsidy, CO2 emission can be maintained at a low level, between 6 and 9 Mtons per year.

4.3. Carbon Intensity of Electricity Generation

Subsidy	2025	2030	2035	2040	2045	2050
CO2 (kg)	8782610	7099560	7273075	6423590	7875520	8931900
Electricity (MWh)	122808.333	157000.000	193898.333	230000.000	264983.333	300527.778
Carbon intensity (kg/MWh)	71.515	45.220	37.510	27.929	29.721	29.721

Table 37: Carbon intensity of electricity generation - Capital Subsidies

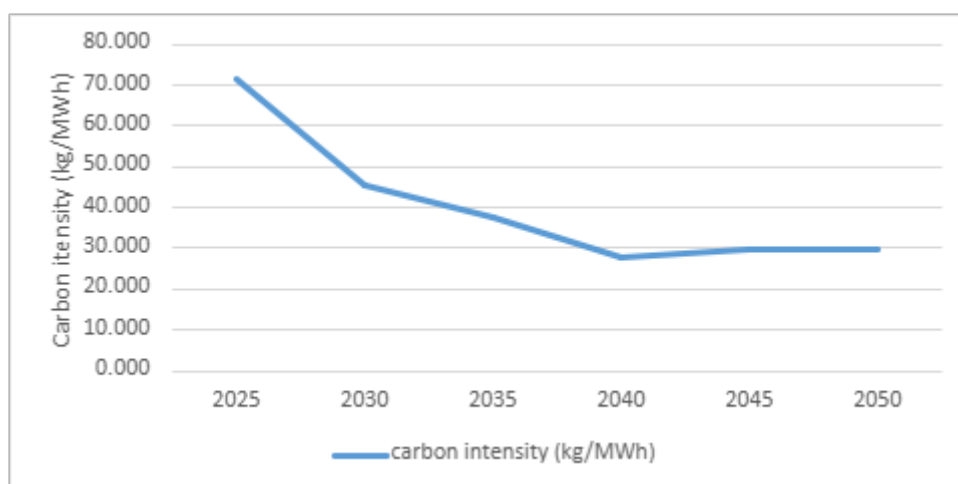


Figure 25: Carbon intensity of electricity generation - Capital Subsidies

Carbon intensity also keeps dropping through the years, in the end it keeps at around 30 kg/MWh from 2045.

4.4. New Electrical Capacities

Subsidy (GW)	2025	2030	2035	2040	2045	2050
ELE_NEW_CCGT				7.14	1.26	1.12
ELE_NEW_WIND_ON	12.44	12.53	12.57	12.4	19.76	23.49
Total	12.44	12.53	12.57	19.54	21.02	24.61

Table 38: New electrical capacities - Capital Subsidies

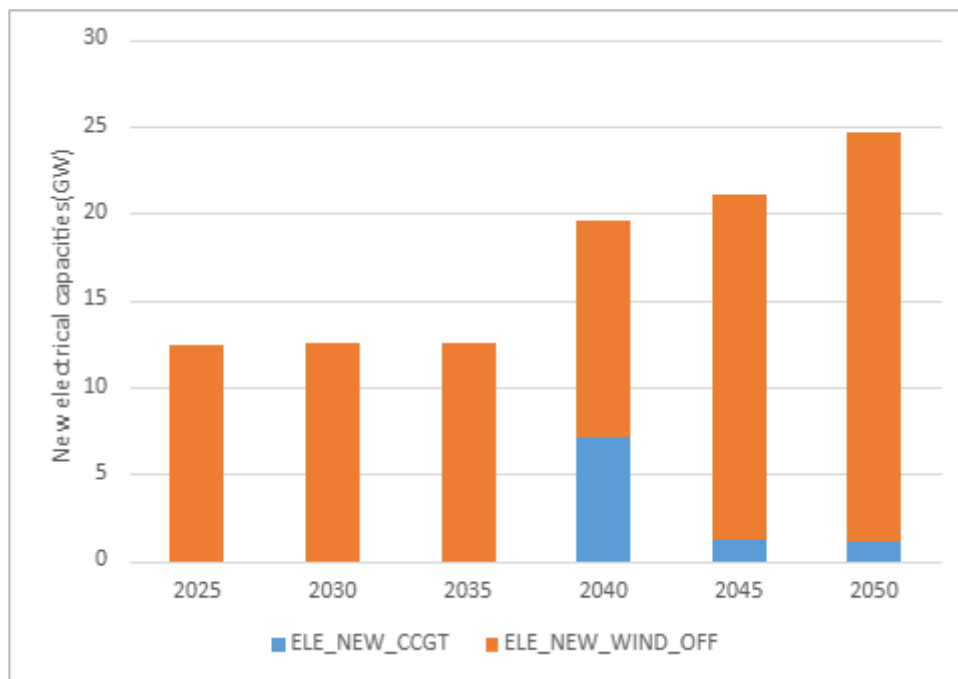


Figure 26: New electrical capacities - Capital Subsidies

Most of the new electrical capacities are offshore wind power. In the years 2040, 2045 and 2050, some new CCGT is installed in order to compensate for the dropping energy share of PV and onshore wind power.

4.5. Full Load Operation Hours of CCGT

Subsidy	2025	2030	2035	2040	2045	2050
Annual activity (PJ)	89.42	72.28	74.053	75.91	93.074	105.55
Capacity (GW)	24.044	24.044	24.044	7.14	8.41	9.54
Full load hours	1033.060	835.043	855.526	2953.237	3074.184	3073.317

Table 39: Full load operation hours of CCGT - Capital Subsidies

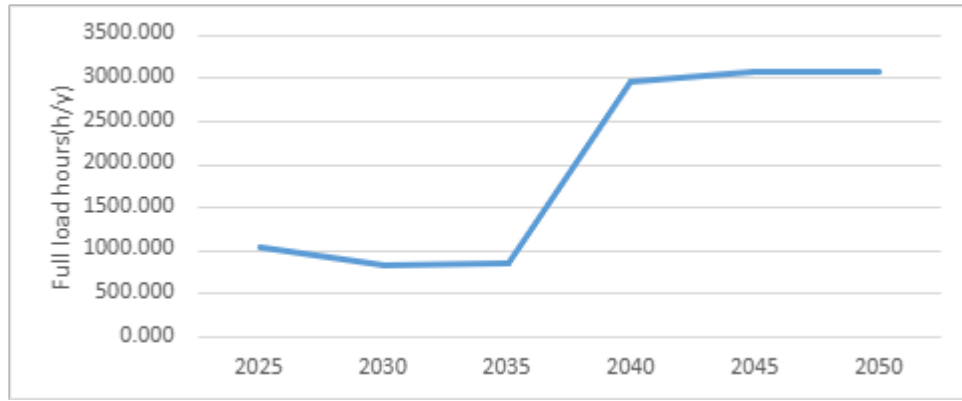


Figure 27: Full load operation hours of CCGT - Capital Subsidies

Full load hours of CCGT slightly decrease first then rise sharply between the years 2035 and 2040, after 2040 it stables at around 3000 hours per year. The sudden rise could be because of the abandoned capacities in 2020.

4.6. RES Curtailment

Subsidy	2025	2030	2035	2040	2045	2050
PV (TWh)	2.701	1.519	1.380	0.984	/	/
PV (%)	12.51%	9.38%	12.79%	18.25%	/	/
Wind OFF (TWh)	1.178	3.194	9.038	18.171	24.572	27.870
Wind OFF (%)	1.79%	2.74%	5.40%	8.35%	9.52%	9.52%
Wind ON total (TWh)	1.211	3.246	3.257	1.963	0	0
Wind ON (%)	8.20%	29.27%	44.10%	53.34%	/	/

Table 40: Renewable Energy Sources Curtailment - Capital Subsidies

From these results, the RES Curtailment under subsidy condition is relatively high, especially for PV and for onshore wind, indicating low energy utilization for these two technologies.

4.7. RES Share in Final Electricity Demand

RES Share	2025	2030	2035	2040	2045	2050
Subsidy	79.15%	86.59%	88.68%	89.36%	88.18%	88.18%

Table 41: Renewable Energy Sources share in final electricity demand - Capital Subsidies

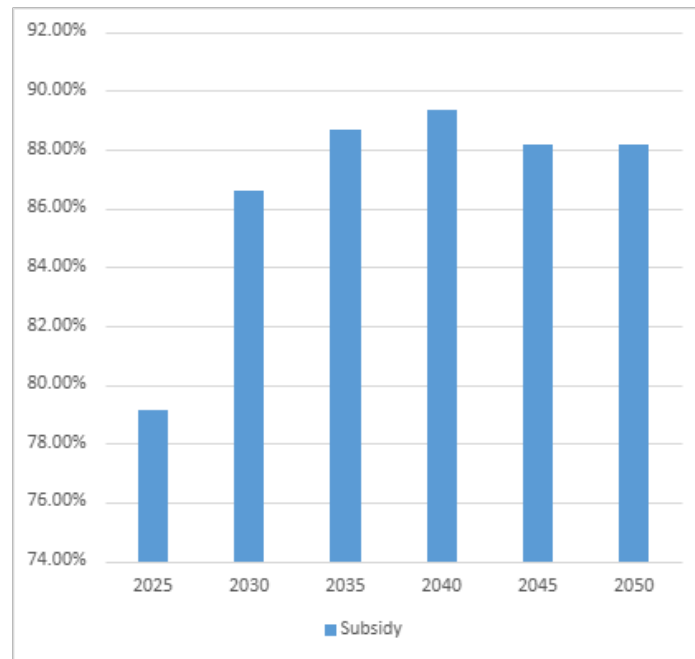


Figure 28: Renewable Energy Sources share in final electricity demand - Capital Subsidies

RES shares will stay at over 70% from 2030. By 2050, RES share arrives at 77.6%. This indicates the success of the subsidy policy.

4.8.RES Share Results on the Time-slice Level

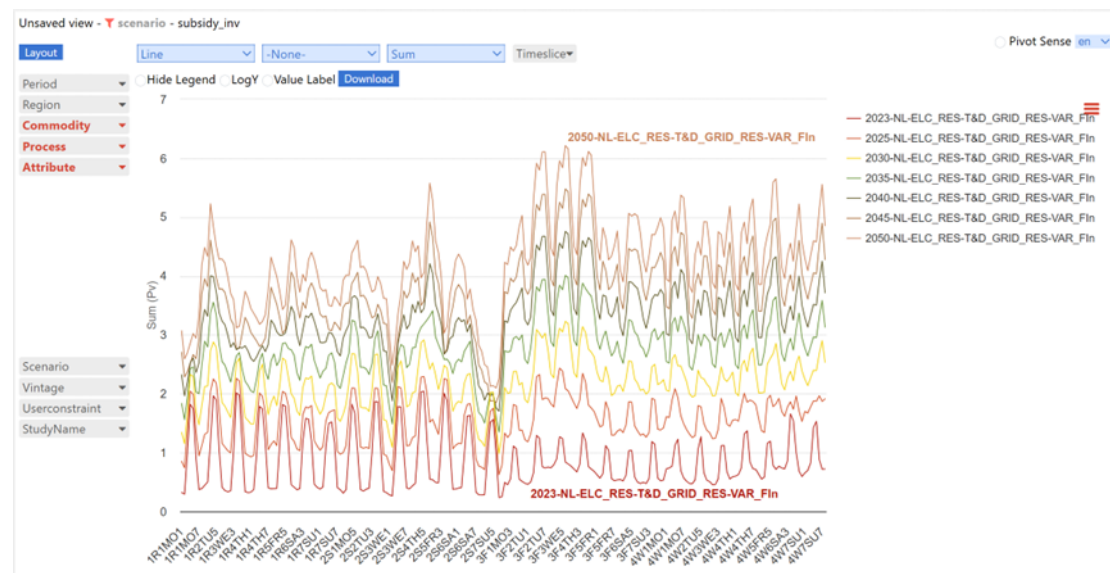


Figure 29: Renewable Energy Sources share results on the time-slice level - Capital Subsidies

4.9. Electricity Generation Cost

Subsidy	Item	2025	2030	2035	2040	2045	2050
CCGT	Cost (MEuro)	659.720	625.450	628.980	832.910	988.080	1120.620
	Electricity (TWh)	24.839	20.078	20.570	21.086	25.854	29.319
	Electricity cost (Euro/MWh)	26.560	31.151	30.577	39.500	38.218	38.221
PV	Cost (MEuro)	205.360	154.022	102.680	51.340	0.000	0.000
	Electricity (TWh)	18.881	14.669	9.406	4.408	0.000	0.000
	Electricity cost (Euro/MWh)	10.877	10.500	10.917	11.646	0.000	0.000
Wind off	Cost (MEuro)	5032.88	10066.330	15113.890	20093.740	23027.580	27432.383
	Electricity (TWh)	64.753	113.431	158.408	199.411	233.658	265.001
	Electricity cost (Euro/MWh)	77.725	88.744	95.411	100.765	98.552	103.518
Wind on	Cost (MEuro)	56.270	42.200	28.130	14.068	0.000	0.000
	Electricity (TWh)	13.558	7.844	4.128	1.717	0.000	0.000
	Electricity cost (Euro/MWh)	4.150	5.380	6.815	8.195	0.000	0.000

Table 42: Electricity generation cost - Capital Subsidies

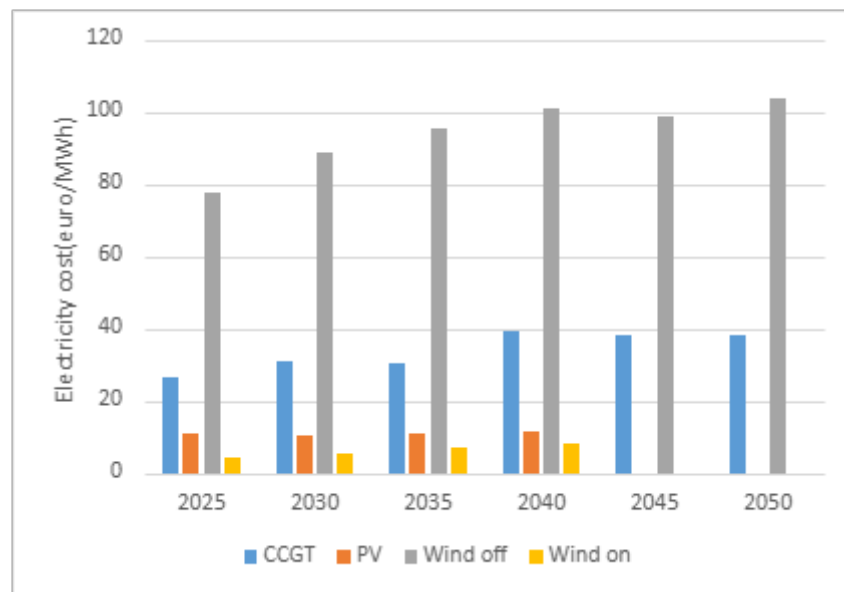


Figure 30: Electricity generation cost - Capital Subsidies

Offshore wind accounts for the highest electricity cost due to its continuously installed new capacities. CCGT is the second highest technology regarding electricity cost per unit.

4.10. Subsidies and Taxes Values

Scenario	2025	2030	2035	2040	2045	2050
Subsidy (€/kW)	-24891.8	-25076.56	-25146.87	-24809.9	-39523.31	-46989.42

Table 43: Subsidies - Capital Subsidies

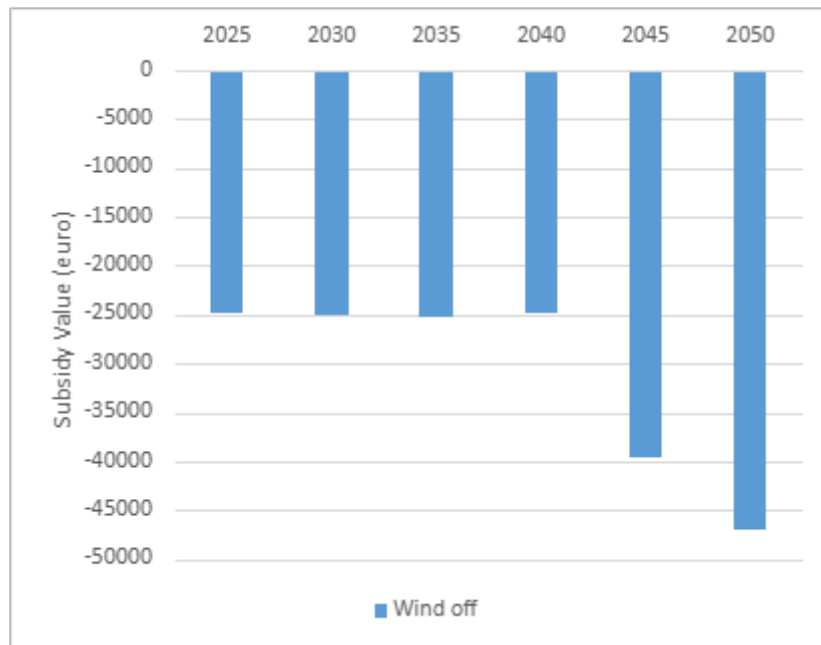


Figure 31: Subsidies - Capital Subsidies

Annual subsidy is around €25000 from 2025 to 2040; it rises obviously from 2045 and arrives at €47000 in 2050.

5. Policy 5: Recycling Carbon Tax Revenues

In the following scenario, the Netherlands implements a CO₂ emissions tax on its power sector using the generated tax revenue to fund renewable electricity consumption subsidies. The policy combines environmental cost internalization, penalizing CO₂ output and promoting market adoption of renewables.

5.1. Electricity Generation Mix [TWh]

The installed capacities of the different existing plants remain the same of the only tax scenario; what changes is new installed GWs of the onshore wind. In fact, after the introduction of the subsidies, a rise is noted and shown in Table 44.

Process	Attribute	2025	2030	2035	2040	2045	2050
CCGT	VAR_Cap (GW)	24.044	24.044	24.044			

PV	VAR_Cap (GW)	20.53	15.4	10.26	5.13		
Existing Wind Off	VAR_Cap (GW)	2.92	2.19	1.46	0.73		
Existing Wind On	VAR_Cap (GW)	5.62	4.22	2.81	1.4		
New Wind On	VAR_Cap (GW)	21.057	46.14	71.92	96.72	116.88	132.56

Table 44: Installed capacity - Recycling Carbon Tax Revenues

To find the electricity generated in TWh, the results are sorted per Var_FOut which provide the values in PJ which are then converted. Table 45 shows the results in TWh.

Process	Attribute	2025	2030	2035	2040	2045	2050
ELC_FIN_DEM	VAR_Fout (TWh)	129.2	176.0	231.1	285.7	331.9	376.4
CCGT	VAR_Fout (TWh)	24.48	18.11	3.86			
PV	VAR_Fout (TWh)	21.94	16.45	10.97	5.48		
Existing Wind Off	VAR_Fout (TWh)	12.53	9.39	6.26	3.13		
Existing Wind On	VAR_Fout (TWh)	14.54	10.91	7.27	3.64		
New Wind On	VAR_Fout (TWh)	54.42	119.2	185.9	250.0	302.1	342.6

Table 45: Electricity generated (in TWh) - Recycling Carbon Tax Revenues

The combined carbon-tax-plus-subsidy scenario leads to a total electricity demand increase from 129 TWh in 2025 to 376 TWh by 2050 whereas the standalone tax results in a range of 123–300 TWh. CCGT generation shows a faster decline in output during this period, since it produces 18 TWh in 2030 instead of 23 TWh of the standalone tax. The existing PVs and onshore/offshore wind facilities operate at higher levels, which results in 10–20 % additional energy output during the mid-2030s compared to the tax-only scenario. Moreover, the deployment of new onshore wind facilities increases rapidly from 54 TWh in 2025 to reach 343 TWh by 2050 which surpasses the 45–260 TWh no subsidies-range and enables the satisfaction of increased power demands while eliminating fossil fuels.

5.2. Overall GHG Emissions

CO₂ emissions from the existing CCGT decrease faster than under a standalone carbon tax. By 2025 the subsidy scenario has already cut emissions to about 8.7 Mt—nearly 19 % lower than the 10.7 Mt seen under the tax-only case. By 2030, emissions under the combined policy fall to roughly 6.4 Mt (a 21 % reduction versus 8.1 Mt), and by 2035 they reach just 1.4 Mt compared with 2.1 Mt under a pure tax. This is mostly due to the stronger displacement of gas-fired generation plants achieved when carbon-tax revenues are reinvested into renewable-electricity subsidies.

Scenario	Commodity	Process	Attribute	2025	2030	2035
1	CO2	CCGT	VAR_Fout (Mt)	10.71684	8.14054	2.06026
5	CO2	CCGT	VAR_Fout (Mt)	8.65723	6.40192	1.36385

Table 46: Overall CO2 emissions

5.3. Carbon Intensity of Electricity Generation

Process	Attribute	2025	2030	2035
CCGT	Carbon intensity (kgCO2/MWh)	36.377	5.901	67.012

Table 47: Carbon intensity of electricity generation - Recycling Carbon Tax Revenues

As shown in Table 47, the carbon intensity of the scenario including the subsidy is way lower than the simple one using just the carbon tax. In the Table, the CO2 emissions are expressed in kg and the energy produced in MWh. The resulting carbon intensity is thus measured in kgCO2/MWh.

5.4. New Electrical Capacities

Process	Attribute	Scenario	2025	2030	2035	2040	2045	2050
New Wind ON	VAR_NCap (GW)	1	17.81	24.2	25.2	23.5	32.9	43.09
New Wind ON	VAR_NCap (GW)	5	21.05	25.08	25.8	24.8	36.3	45.64

Table 48: New electrical capacities

The revenue-recycling scenario (Scenario 5) produces higher new onshore wind capacity than the carbon-tax-only case. The revenue-recycling scenario (Scenario 5) will deploy 21.1 GW of new onshore wind capacity in 2025 which exceeds the 17.8 GW installed in the tax-only case. The difference between the two scenarios decreases to 0.9 GW during 2030 (25.1 GW vs. 24.2 GW) before shrinking to 0.6 GW in 2035. The difference between the two scenarios increases starting from 2040 until 2050 when Scenario 5 adds 1.3 GW in 2040 and 3.4 GW in 2045 before reaching a 2.6 GW advantage in 2050, with 45.6 GW compared to the 43.1 GW of the only tax scenario.

5.5. Full Load Operation Hours of CCGT

As previously done, to find the full operation hours a computation is needed. Particularly, after extracting for the CCGT process the VAR_Cap (capacity, in GW) and Var_FOut (generation, in TWh) attributes, by dividing them it is possible to find the full-load hours per year. A conversion from GW to TW was needed for the VAR_Cap results.

The full-load hours are shown in Table 49.

Process	Attribute	2025	2030	2035
CCGT	VAR_Cap (TW)	0.024044	0.024044	0.024044
	VAR_Fout (TWh)	24.48333333	18.10555556	3.85555556
	full-load hours	1018.272057	753.0176159	160.3541655

Table 49: Full load operation hours of CCGT - Recycling Carbon Tax Revenues

5.6. RES Curtailment

RES Curtailment (TWh)	2025	2030	2035	2040	2045	2050
PVs	0	0	0	0		
PVs (%)	0	0	0	0		
Offshore Wind	0	0	0	0		
Offshore Wind (%)	0	0	0	0		
Onshore Wind	1.15	2.20	3.25	4.26	5.10	5.78
Onshore Wind (%)	1.63%	1.66%	1.65%	1.65%	1.66%	1.66%

Table 50: Renewable Energy Sources Curtailment - Recycling Carbon Tax Revenues

Table 50 shows how PVs and offshore wind present no curtailed energy, meaning that all the produced energy is actually utilized and delivered to the grid. On the other hand, on shore wind present higher values, which are however acceptable and in percentage constant over the years.

5.7. RES Share in Final Electricity Demand

Scenario	Commodity	Process	Attribute	2025	2030	2035	2040	2045	2050
1	ELC_FIN	ELC_FIN_DEM	VAR_Fout (TWh)	122.81	157.00	193.90	230.00	264.98	300.53
	ELC_GRID	T&D_GRID	VAR_Fout (TWh)	31.60	24.99	24.94	28.39	35.48	40.24
	ELC_GRID_RES	T&D_GRID_RES	VAR_Fout (TWh)	91.21	132.01	168.95	201.61	229.51	260.29
5	ELC_FIN	ELC_FIN_DEM	VAR_Fout (TWh)	129.19	175.99	231.10	285.68	331.89	376.41
	ELC_GRID	T&D_GRID	VAR_Fout (TWh)	25.76	19.99	20.72	23.46	29.82	33.82
	ELC_GRID_RES	T&D_GRID_RES	VAR_Fout (TWh)	103.43	156.00	210.38	262.22	302.07	342.59

Table 51: Renewable Energy Sources share in final electricity demand

The RES share reaches 80 % in 2025 (103.4 TWh of output versus 129.2 TWh demand) after carbon-tax revenues are used for renewable-electricity subsidies and then accelerates more sharply to reach nearly 89 % by 2030 and exceed 91 % by 2035. The renewable energy supply maintains a consistent level of 91–92 % of final demand throughout the mid-century period (e.g., 262.2 TWh of RES against 285.7 TWh total in 2040, and 342.6 TWh out of 376.4 TWh

in 2050). The subsidy regime drives both increased domestic clean energy production together with higher overall consumption.

5.8. Electricity Generation Cost

Baseline	Item	2025	2030	2035	2040	2045	2050
CCGT	Cost (MEuro)	657.17	611.24	508.65			
	Electricity (TWh)	24.48	18.10	3.85			
	Electricity cost (Euro/MWh)	26.84	33.76	131.93			
PV	Cost (MEuro)	205.36	154.02	102.68	51.34		
	Electricity (TWh)	21.94	16.45	10.97	5.48		
	Electricity cost (Euro/MWh)	9.36	9.36	9.36	9.36		
Wind off	Cost (MEuro)	29.27	21.95	14.63	7.31		
	Electricity (TWh)	12.53	9.39	6.26	3.13		
	Electricity cost (Euro/MWh)	2.34	2.34	2.34	2.33		
Wind on	Cost (MEuro)	3362.5	7286.81	11321	15201	17585	20814
	Electricity (TWh)	68.96	130.15	193.14	253.60	302.07	342.59
	Electricity cost (Euro/MWh)	48.76	55.99	58.61	59.94	58.22	60.75

Table 52: Electricity generation cost - Recycling Carbon Tax Revenues

The results shown in Table 52 enhance how the most cost-effective technology is offshore wind, while the most expensive is the onshore wind, due to the new capacity installed.

5.9. Subsidies and Taxes Values

All the costs are expressed in M€.

Scenario	Attribute	2023	2025	2030	2035	2040	2045	2050
1	Tax Revenues	1308.0047	842.037	814.054	309.039			
5	Tax Revenues	1308.0047	680.21	640.19	204.57			
	Subsidies expenditure	-388.14	-744.68	-1123.2	-1514.72	-1887.99	-2174.91	-2466.64

Table 53: Subsidies and taxes value

Table 53 shows how the tax earned from CO₂ is more profitable in the first Scenario (tax only), where the revenues achieved were roughly 20 to 34% higher during the years than Scenario 5. In fact, Scenario 5 shows lower revenues from the taxation, and from 2025 even more expenses

for the subsidies than returns. In this case, to compensate the revenues and expenses, two possible solutions would be to either increase the tax or diminish the subsidies.

6. Policy 6: RES Share in Annual Electricity Consumption

6.1. Electricity Generation Mix [TWh]

RES50%(TWh)	2025	2030	2035	2040	2045	2050
CCGT	42.472	77.922	94.753	110.817	126.994	144.031
PV	21.019	16.453	10.967	5.483	0.000	0.000
Wind Off	12.131	9.333	6.261	3.131	0.000	0.000
Wind ON	47.183	52.711	79.714	106.383	132.492	150.264

Table 54: Electricity generation mix - 50% RES Share in Annual Electricity Consumption

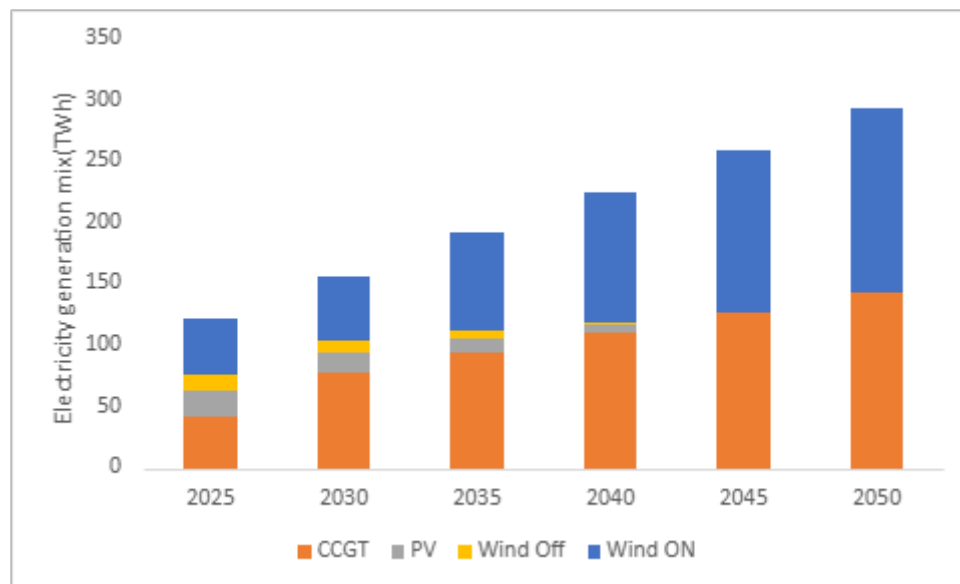


Figure 32: Electricity generation mix - 50% RES Share in Annual Electricity Consumption

With the constraint of 50% of electricity coming from RES, both onshore wind power and CCGT keep growing throughout the years. These two technologies keep squeezing other technologies share. From 2045, only onshore wind and CCGT exist on the market with the share close to 50% - 50%.

6.2. Overall GHG Emissions

RES50% (Mtons)	2025	2030	2035	2040	2045	2050
CO2 emission	15.017	27.552	33.502	33.757	38.685	43.874

Table 55: Overall CO2 emissions - 50% RES Share in Annual Electricity Consumption

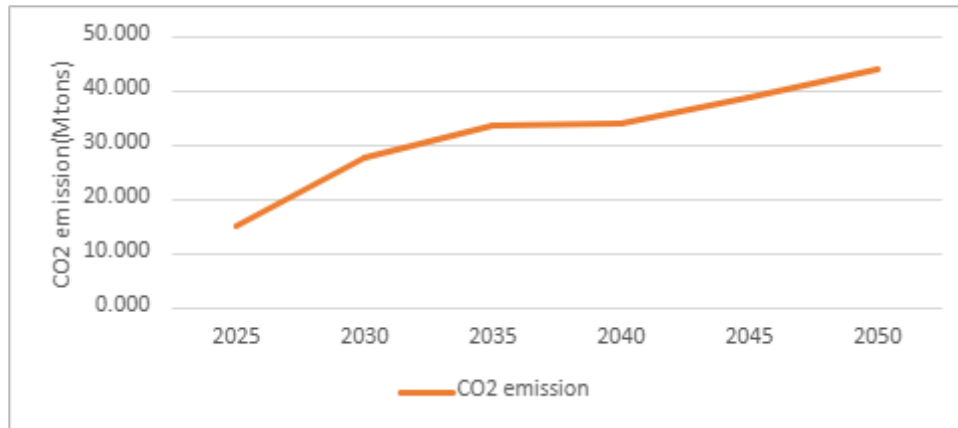


Figure 33: Overall CO2 emissions - 50% RES Share in Annual Electricity Consumption

CO2 emissions keep growing with the generation of CCGT. By 2050, around 44 Mton of CO2 is emissions yearly.

6.3. Carbon Intensity of Electricity Generation

RES50%	2025	2030	2035	2040	2045	2050
CO2 (kg)	15017180	27551670	33502420	33757260	38685110	43874130
Electricity (MWh)	122808.333	157000.000	193977.778	230086.111	265075.000	300633.333
Carbon intensity (kg/MWh)	122.281	175.488	172.713	146.716	145.940	145.939

Table 56: Carbon intensity of electricity generation - 50% RES Share in Annual Electricity Consumption

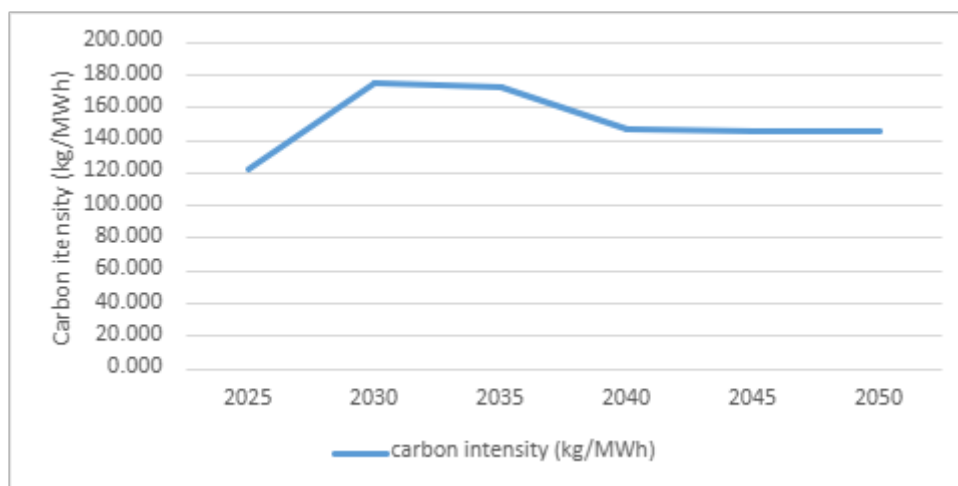


Figure 34: Carbon intensity of electricity generation - 50% RES Share in Annual Electricity Consumption

Carbon intensity grows first to 175 kg/MWh till 2030, then declines and eventually stables at 146 kg/MWh by 2050. This indicates that by controlling RES share, carbon intensity can be limited.

6.4. New Electrical Capacities

RES50% (GW)	2025	2030	2035	2040	2045	2050
ELE_NEW_CCGT	/	/	/	18.22	3.052	2.85
ELE_NEW_WIND_ON	12.79	3.39	11.85	11.72	21.33	13.23
Total	12.79	3.39	11.85	29.94	24.382	16.08

Table 57: New Electrical Capacities - 50% RES Share in Annual Electricity Consumption

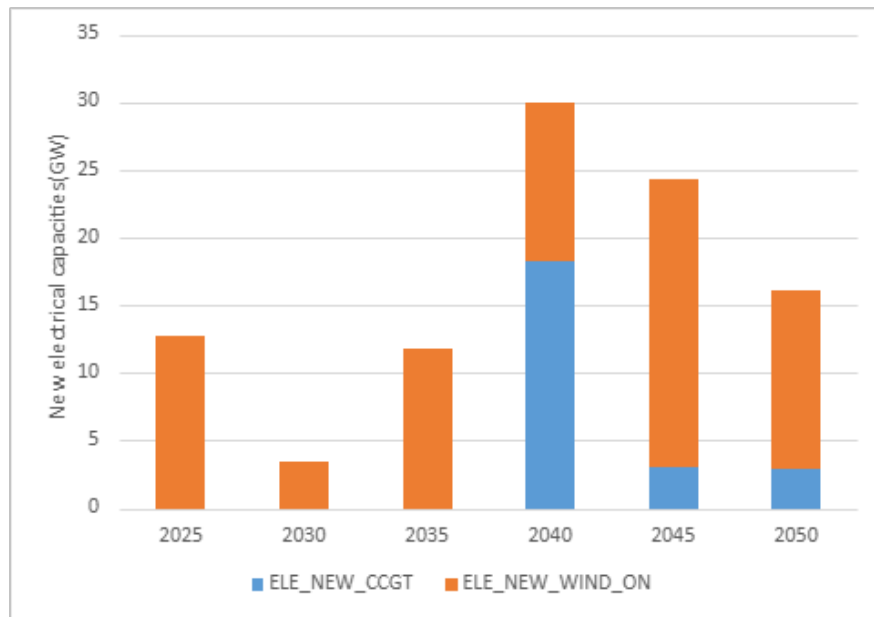


Figure 35: New Electrical Capacities - 50% RES Share in Annual Electricity Consumption

New onshore wind is installed every year to meet the growing demand while meeting the RES share constraint. From 2040, new CCGTs also be installed.

6.5. Full Load Operation Hours of CCGT

RES50%	2025	2030	2035	2040	2045	2050
Annual activity (PJ)	152.9	280.52	341.11	398.94	457.18	518.51
Capacity (GW)	24.044	24.044	24.044	18.22	21.27	24.13
Full load hours	1766.43 7	3240.81 8	3940.80 8	6082.14 4	5970.59 0	5968.94 1

Table 58: Full load operation hours of CCGT - 50% RES Share in Annual Electricity Consumption

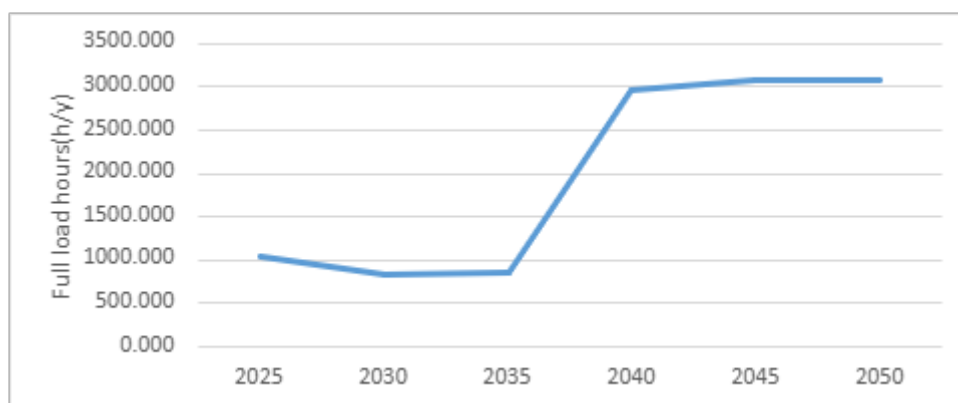


Figure 36: Full load operation hours of CCGT - 50% RES Share in Annual Electricity Consumption

Full load hours of CCGT slightly decrease in 2030, then rise sharply in 2040, and eventually stables at around 6000 hours by 2050.

6.6. RES Curtailment

RES50%	2025	2030	2035	2040	2045	2050
PV (TWh)	0.562	0	0	0		
PV (%)	2.60%	0%	0%	0%		
Wind OFF (TWh)	0.403	0.067	0.006	0.003	0.000	0.000
Wind OFF (%)	3.22%	0.71%	0.09%	0.09%	/	/
Wind ON total (TWh)	1.198	0.900	1.339	1.759	2.220	2.528
Wind ON (%)	2.48%	1.68%	1.65%	1.63%	1.65%	1.65%

Table 59: Renewable Energy Sources Curtailment - 50% RES Share in Annual Electricity Consumption

As can be seen from the table, all RES have a very low curtailment, indicating full utilization of RES power plants.

6.7. RES Share in Final Electricity Demand

RES Share	2025	2030	2035	2040	2045	2050
RES50%	65.42%	50.00%	49.98%	49.98%	49.98%	49.98%

Table 60: Renewable Energy Sources share in final electricity demand - 50% RES Share in Annual Electricity Consumption

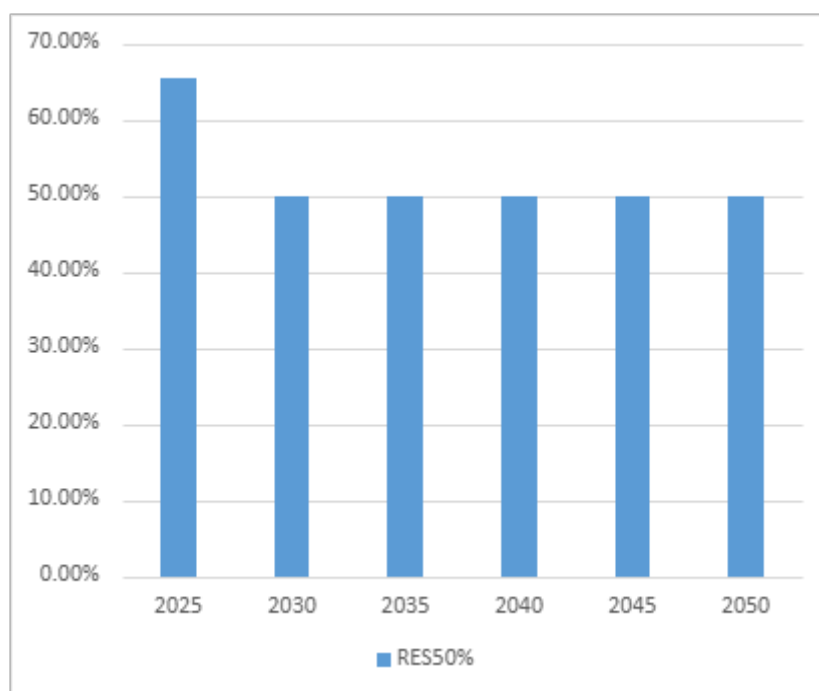


Figure 37: Renewable Energy Sources share in final electricity demand - 50% RES Share in Annual Electricity Consumption

RES share in final electricity demand is stabilized at around 50% from the year 2030. The value is slightly less than 50% possibly because of the imported electricity share.

6.8. RES Share Results on the Time-slice Level

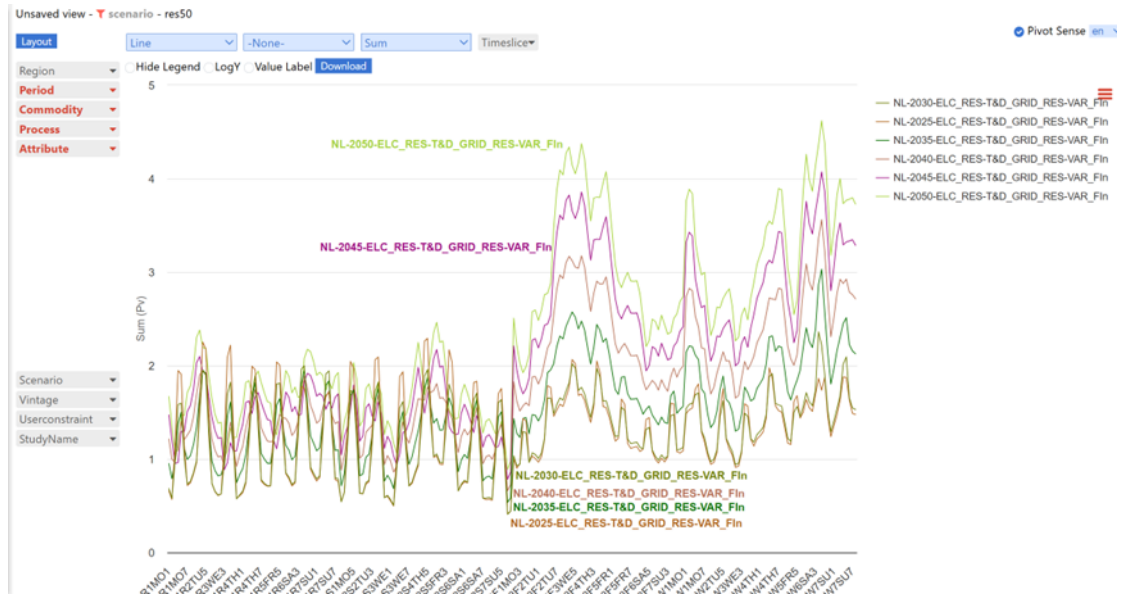


Figure 38: Renewable Energy Sources share results on the time-slice level - 50% RES Share in Annual Electricity Consumption

6.9. Electricity Generation Cost

RES50%	Item	2025	2030	2035	2040	2045	2050
CCGT	Cost (MEuro)	786.680	1041.932	1163.110	2534.260	2941.520	3336.085
	Electricity (TWh)	42.472	77.922	94.753	110.817	126.994	144.031
	Electricity cost (Euro/MWh)	18.522	13.371	12.275	22.869	23.163	23.162
PV	Cost (MEuro)	205.360	154.022	102.680	51.340	0.000	0.000
	Electricity (TWh)	21.019	16.453	10.967	5.483	0.000	0.000
	Electricity cost (Euro/MWh)	9.770	9.361	9.363	9.363	0.000	0.000
Wind off	Cost (MEuro)	29.270	21.950	14.630	7.310	0.000	0.000
	Electricity (TWh)	12.131	9.333	6.261	3.131	0.000	0.000
	Electricity cost (Euro/MWh)	2.413	2.352	2.337	2.335	0.000	0.000
Wind on	Cost (MEuro)	2064.680	2582.900	4429.470	6256.438	7583.769	9129.250
	Electricity (TWh)	47.183	52.711	79.714	106.383	132.492	150.264
	Electricity cost (Euro/MWh)	43.759	49.001	55.567	58.810	57.240	60.755

Table 61: Electricity generation cost - 50% RES Share in Annual Electricity Consumption

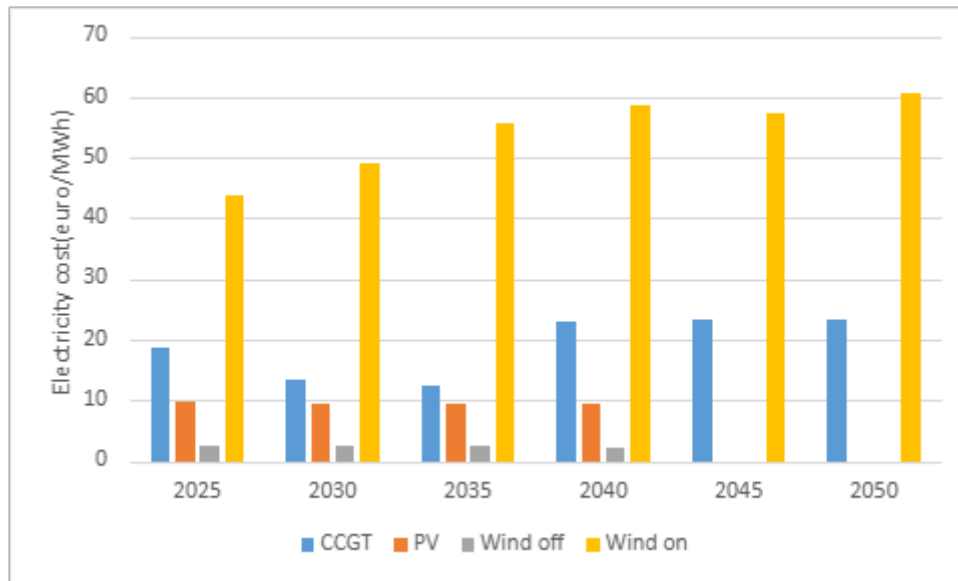


Figure 39: Electricity generation cost - 50% RES Share in Annual Electricity Consumption

Onshore wind accounts for the most expensive technology generating electricity per MWh, achieving 60 euro/MWh, followed by CCGT at 23 euro/MWh. This is because of the newly installed capacities of onshore wind and CCGT throughout the years.

7. Policy 7: 24/7 Carbon-Free Energy A: 50%

7.1. Electricity Generation Mix [TWh]

As shown in the table below, Combined Cycle Gas Turbines (CCGT) dominate the electric system, increasingly growing from 42.5 TWh in 2025 to 242.4 TWh in 2050. This indicates that the model uses dispatchable fossil capacity to ensure supply within each time-slice during the 24/7 constraint.

Offshore Wind as well as PV gradually disappear within the mix. Offshore wind follows along that same path, and PV drops down to zero by 2045. Integrating these variable sources lacks flexibility with storage, likely resulting from high curtailment levels.

Onshore Wind does maintain a presence all throughout that period peaking at 79 TWh in 2035, although facing a small decline of 58 TWh in 2050.

In general, the policy was intended for ensuring 50% carbon-free electricity, however, the mix becomes increasingly carbon-intensive along time, which suggests the policy constraint was either too weakly implemented or infeasible without firm low-carbon options.

Policy 7a (TWh)	2025	2030	2035	2040	2045	2050
CCGT	42.473	78.500	96.803	145.374	213.589	242.394
PV	20.796	10.366	10.969	5.485	0.000	0.000
Wind Off	12.232	4.819	6.264	3.132	0.000	0.000
Wind ON	47.310	64.046	79.879	76.244	51.545	58.268

Table 62: Electricity generation mix - 24/7 50% Carbon-Free Energy

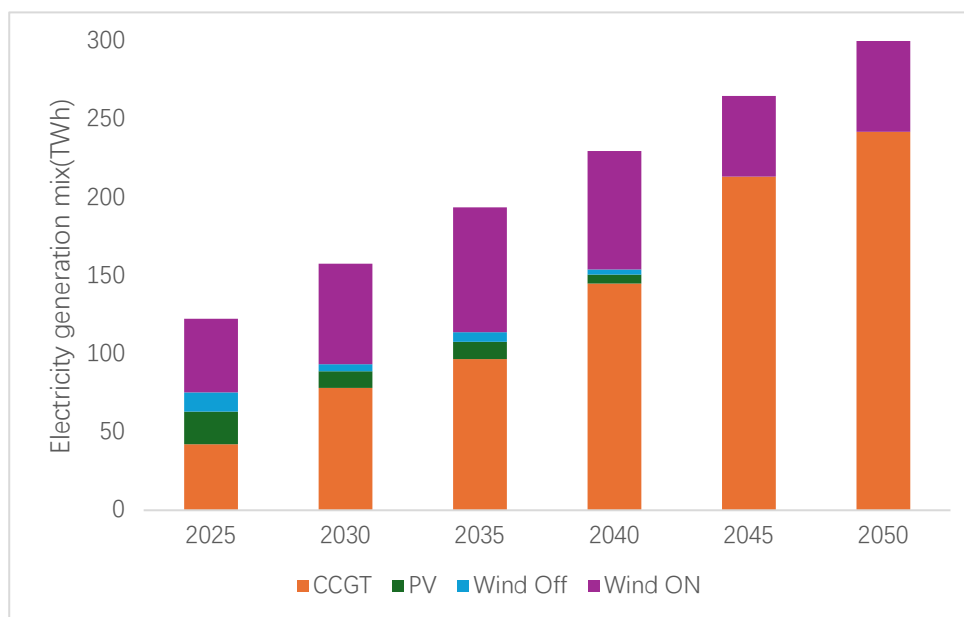


Figure 40: Electricity generation mix - 24/7 50% Carbon-Free Energy

7.2. Overall GHG Emissions

CO₂ emissions in this policy show a steady increase over time because it increased from 15Mt back in 2025 to nearly 74Mt by 2050. This growth in gas-fired electricity aligns with this trend as renewable options decline, making it dominant.

Since the policy aims for 50% carbon-free electricity in every time-slice, this increase suggests that its actual implementation fails for deeply decarbonizing. The system prioritizes reliability from fossil-based generation instead and the carbon footprint increased during this time.

Policy 7A (Mtons)	2025	2030	2035	2040	2045	2050
CO ₂ emission	15.01718	27.75536	34.22663	44.28323	65.06243	73.837

Table 63: Overall CO₂ emissions - 24/7 50% Carbon-Free Energy

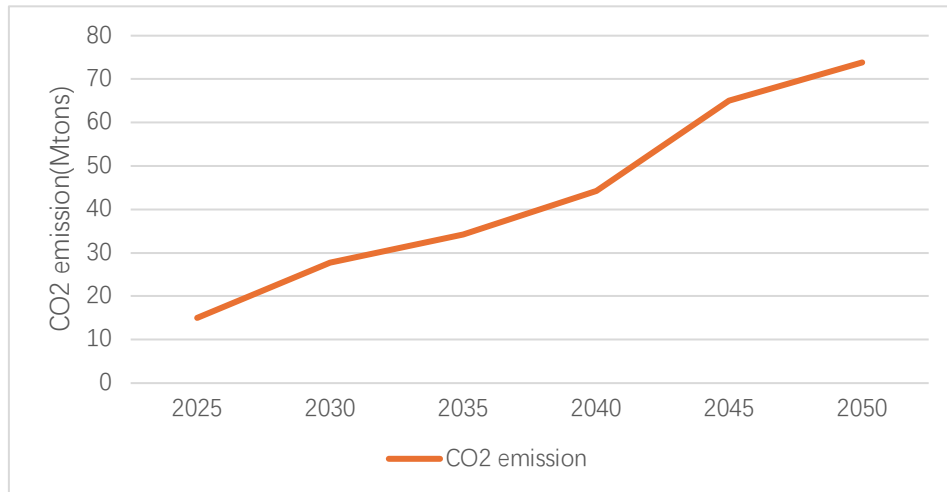


Figure 41: Overall CO2 emissions - 24/7 50% Carbon-Free Energy

7.3. Carbon Intensity of Electricity Generation

The carbon intensity of electricity generation under Policy 7a increases intensively from 2025 to 2050, from 122 kg CO₂/MWh in 2025 to around 246 kg CO₂/MWh in 2050. Therefore, the gradual increase in carbon intensity demonstrates the fossil-based (mostly CCGT based) share of generation which increases, given that renewables, particularly PV and offshore wind, are being displaced by fossil-based generation.

The system becomes more carbon intense over time despite the policy's intended goal of providing at least 50% carbon-free electricity at all times.

Policy_7a	2025	2030	2035	2040	2045	2050
CO2 (kg)	15017184.23	27755357.14	34226629.52	44283227.74	65062432.52	73837000.02
Electricity (MWh)	122810.7306	157000	193898.4018	230000	264986.0731	300529.9087
Carbon intensity (kg/MWh)	122.2790888	176.7857143	176.5183684	192.5357728	245.5315171	245.689357

Table 64: Carbon intensity of electricity generation - 24/7 50% Carbon-Free Energy

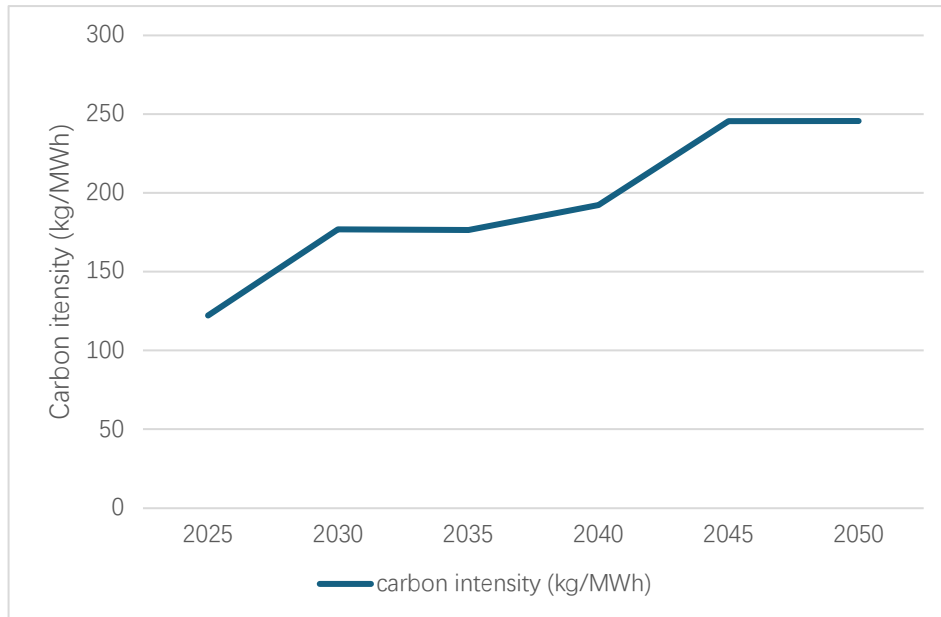


Figure 42: Carbon intensity of electricity generation - 24/7 50% Carbon-Free Energy

7.4. New Electrical Capacities

Onshore wind (ELE_NEW_WIND_ON) dominates in the early years with new capacity growing from 12.8 GW in 2025 (and 15.3 GW in 2030 to a peak value in 2030) but then becomes almost nonexistent until a final peak in 2050 of 20.9 GW.

New CCGT capacity (ELE_NEW_CCGT) growth follows a different pattern with a peak of 21.25 GW in 2040. This represents a continued reliance on dispatchable fossil capacity to provide reliability, as variable renewables (PV, offshore wind) use is diminished or phased out in these years

The last peak from wind in 2050, may show a last effort to balance the system, but the overall results show a preference in gas investments over renewables.

Policy 7A (GW)		2025	2030	2035	2040	2045	2050
ELE_NEW_CCGT	VAR_Ncap				21.25	8.33	4.36
ELE_NEW_WIND_ON	VAR_Ncap	12.79	15.3			1.67	20.87

Table 65: New electrical capacities - 24/7 50% Carbon-Free Energy

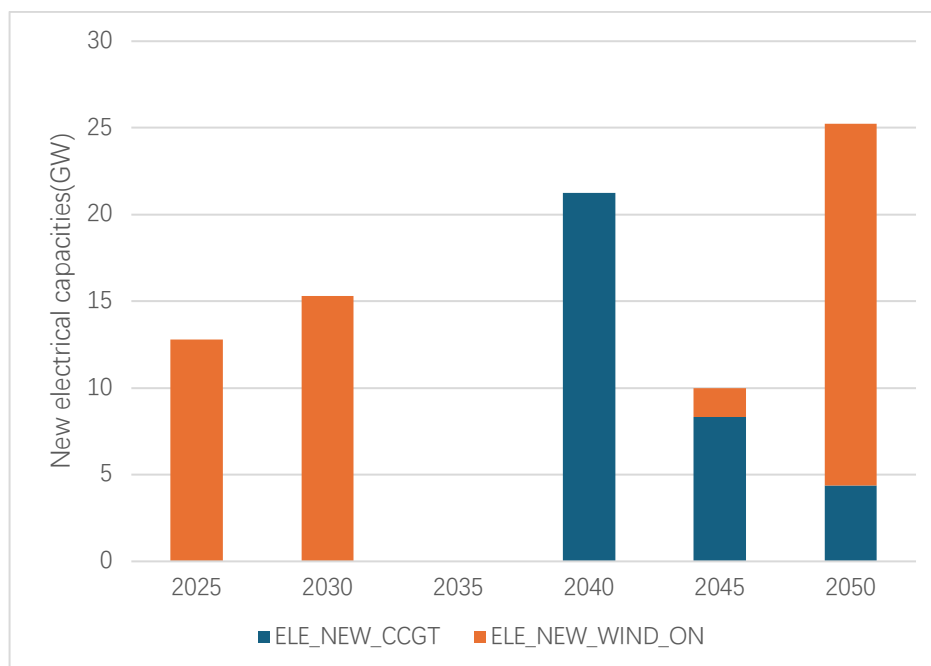


Figure 43: New electrical capacities - 24/7 50% Carbon-Free Energy

7.5. Full Load Operation Hours of CCGT

The CCGT full load hours increase significantly over the simulation horizon such that full load hours increase from 1,766 hours in 2025 to over 7,200 hours by 2045. In general this indicates reliance on CCGT as the main dispatchable resource of the system.

The large increase after 2035 shows a path away from the renewables and towards gas as the central resource.

Policy_7a	2025	2030	2035	2040	2045	2050
Annual activity (PJ)	152.9	282.6	348.48	523.34	768.91	872.61
Capacity (GW)	24.044	24.044	24.044	21.25	29.58	33.94
Full load hours	1766.437457	3264.848	4025.952	6841.046	7220.626	7141.77

Table 66: Full load operation hours of CCGT - 24/7 50% Carbon-Free Energy

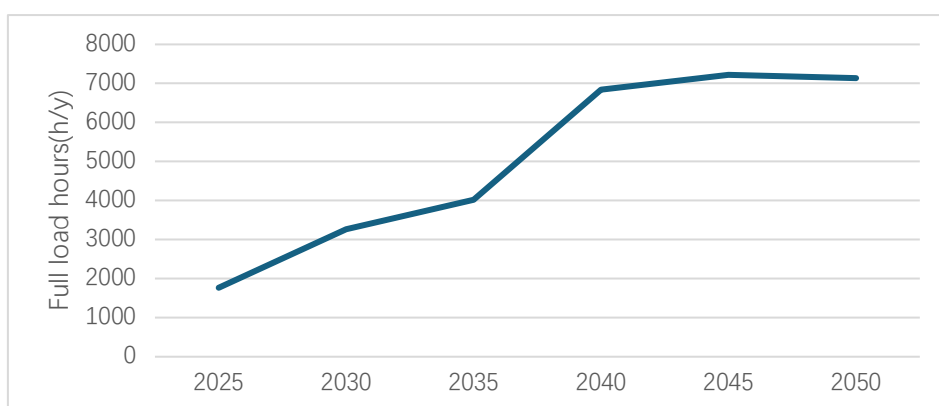


Figure 44: Full load operation hours of CCGT - 24/7 50% Carbon-Free Energy

7.6. RES Curtailment

Under Policy 7a, curtailment peaks in 2030, particularly for PV (36%) and offshore wind (49%), driven by potential overcapacity or lack of grid flexibility. After 2035, curtailment is reduced to approximately zero for all RES technologies, showing better system integration and matching of RES generation with demand. Overall, early years show significant curtailment even as the policy leads with efficient RES in the long run.

Policy_7a	2025	2030	2035	2040	2045	2050
PV (TWh)	0.785068	5.822249	0	0	0	0
PV (%)	3.64%	35.97%	0%	0%	0%	0%
Wind OFF (TWh)	0.301564	4.580933	0.003318	0.001659	0	0
Wind OFF (%)	2.41%	48.73%	0.05%	0.05%	0%	0%
Wind ON total (TWh)	1.071905	20.87822	1.33903	1.269268	0.857716	0.967308
Wind ON (%)	2.22%	24.58%	1.65%	1.64%	1.64%	1.63%

Table 67: Renewable Energy Sources Curtailment - 24/7 50% Carbon-Free Energy

7.7. RES Share in Final Electricity Demand

The proportion of renewable energy in final electricity demand appears strong initiating at 59% in 2025 and holding steady to peak at a little higher than this at 63.8% in 2030, which exceeds the 50% target particularly impressive given that the annual average has included coal generated electricity. Although its performance remains solid until 2035, RES is declining rapidly finishing at 19.35% at 2050.

The steady decline of RES share throughout the reporting period shows that the 50% carbon-free electricity 24/7 policy objective could not be sustained throughout the time-frame.

	2025	2030	2035	2040	2045	2050
policy_7a	59.04%	63.83%	39.54%	28.34%	19.52%	19.35%

Table 68: Renewable Energy Sources share in final electricity demand - 24/7 50% Carbon-Free Energy

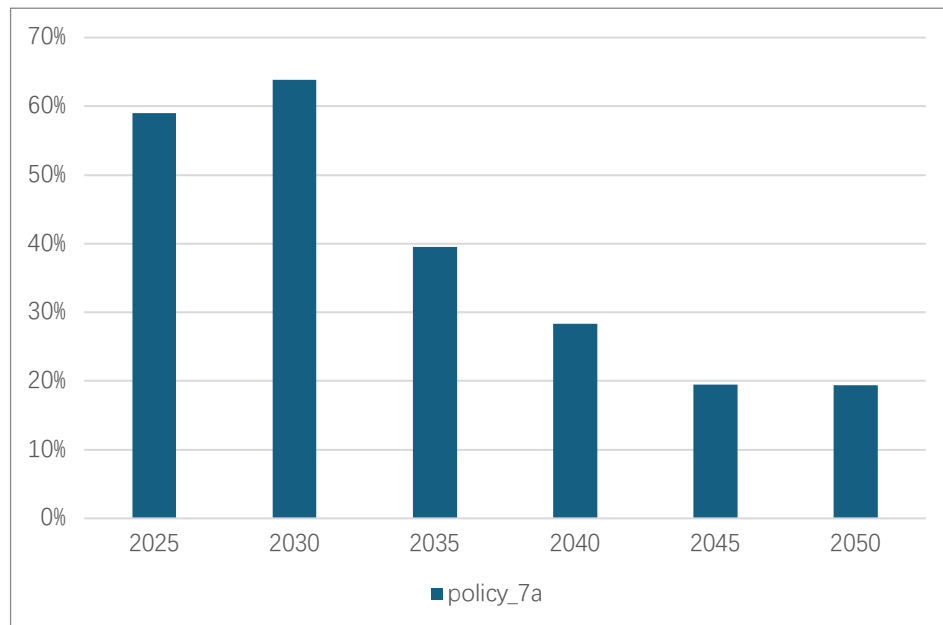


Figure 45: Renewable Energy Sources share in final electricity demand - 24/7 50% Carbon-Free Energy

7.8. RES Share Results on the Time-slice Level

The time-slice analysis suggests that the 50% carbon-free electricity share is not consistently achieved during the course of the year. In the case of 2025, the time-slices taken for the analysis show there are time periods throughout the year when VAR_FIn exceeds, or is near, half of the demand.

When looking at 2035, 2040, especially 2050 the RES share is very variable, and is predominantly below, 50%, in most time-slices, waiting to a point where share fluctuated between 20% and 40%. This indicates that, the system does not supply clean electricity evenly across all hours and across seasons, which is antithetical to the nature of the 24/7 policy target.

The decline in staring, hour performance was strong between the years due to:

- the amount of PV and offshore wind capacity is much lower,
- the extremely high level of curtailment,
- growth of CCGT in the electricity supply in the mix.

Overall, the time-slice view exposes that the 24/7 carbon-free target in Policy 7a had not been structurally enforced, and the system increasingly drew from fossil sources of electricity to meet demand in those times when a supply of renewables could not be available.

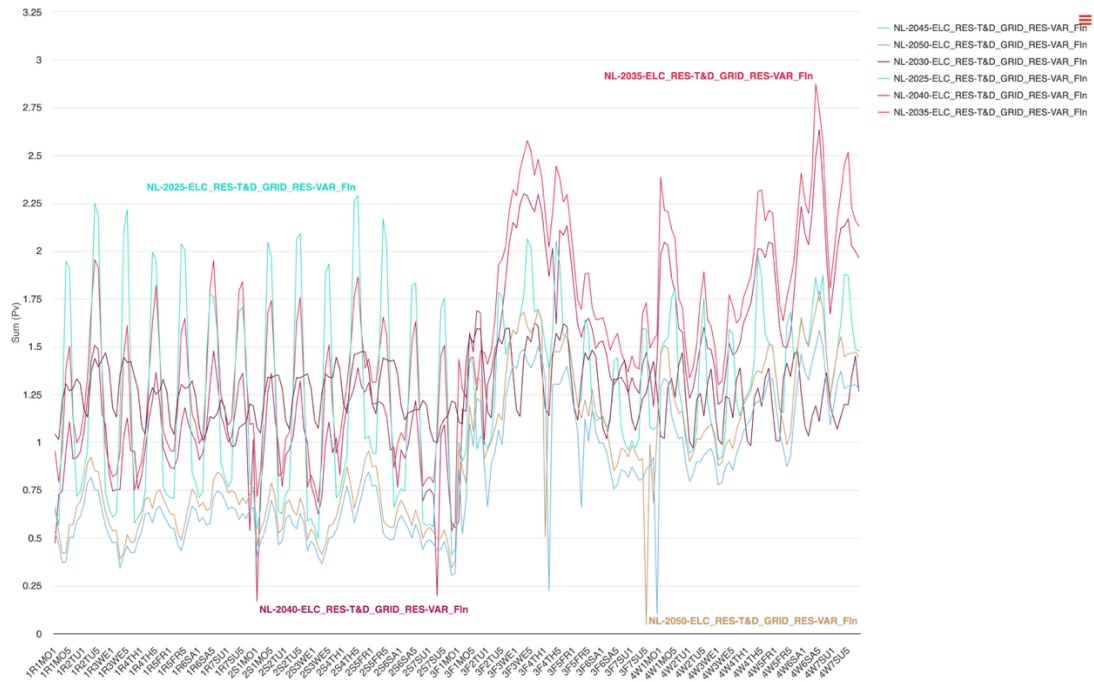


Figure 46: Renewable Energy Sources share results on the time-slice level - 24/7 50% Carbon-Free Energy

7.9. Electricity Generation Cost

Generation costs vary significantly among electricity generation technologies. Wind On has remained one of the most expensive technologies since its cost ranges between €43/MWh and €61/MWh. This probable cost factor restrained its deployment despite its high potential. The cost remains comparatively stable in time, being somewhat higher than prices from other technologies. CCGT is the cheapest option for most years, especially in 2030 and 2035, where its cost drops below €14/MWh, although it rises again after 2040, it remains cost-competitive throughout time. PV and Wind Off are cheap in early years, but are phased out completely by 2045.

Policy7a	Item	2025	2030	2035	2040	2045	2050
CCGT	Cost (MEuro)	786.680	967.320	1291.660	3071.164	4355.930	4978.940
	Electricity (TWh)	42.473	78.500	96.803	145.374	213.589	242.394
	Electricity cost (Euro/MWh)	18.522	12.323	13.343	21.126	20.394	20.541
PV	Cost (MEuro)	205.360	154.022	102.680	51.340	0.000	0.000
	Electricity (TWh)	20.796	10.366	10.969	5.485	0.000	0.000
	Electricity cost (Euro/MWh)	9.875	14.858	9.361	9.361	0.000	0.000
Wind off	Cost (MEuro)	29.270	21.950	14.630	7.310	0.000	0.000
	Electricity (TWh)	12.232	4.819	6.264	3.132	0.000	0.000
	Electricity cost (Euro/MWh)	2.393	4.554	2.336	2.334	0.000	0.000
Wind on	Cost (MEuro)	2064.680	4453.440	4439.370	4425.308	2665.797	3540.020
	Electricity (TWh)	47.310	64.046	79.879	76.244	51.545	58.268
	Electricity cost (Euro/MWh)	43.642	69.535	55.576	58.042	51.718	60.754

Table 69: Electricity generation cost - 24/7 50% Carbon-Free Energy

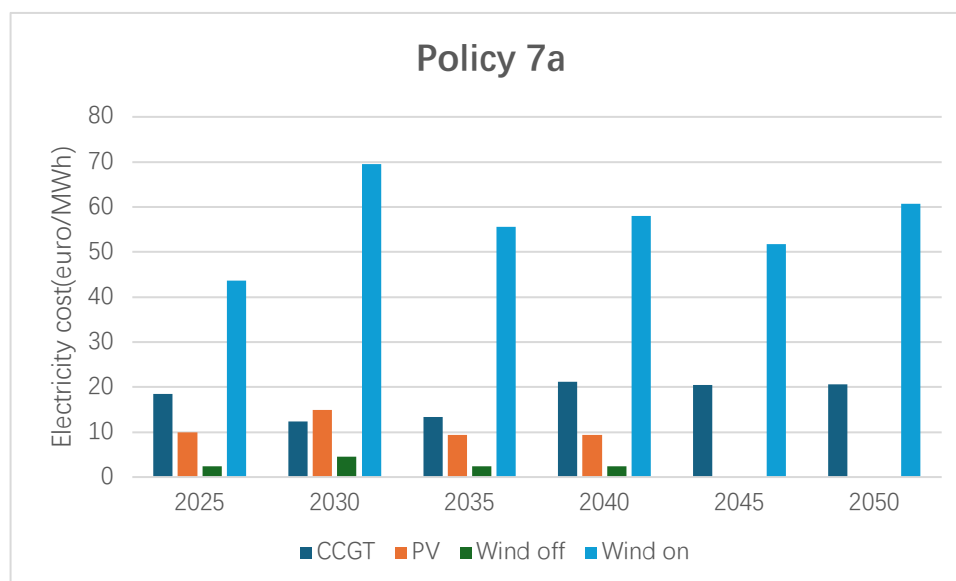


Figure 47: Electricity generation cost - 24/7 50% Carbon-Free Energy

8. Policy 7: 24/7 Carbon-Free Energy A: 90%

8.1. Electricity Generation Mix [TWh]

Compared to Policy 7a, Policy 7b has a much larger proportion of renewable energy throughout the time horizon, but particularly from 2030-2045. Onshore wind is the primary source of generation and grows rapidly, from 47.3 TWh in 2025 to 177.5 TWh in 2040, and stays above 160 TWh/year through 2045. Similarly, the CCGT remains low until 2040, starting at 42.5 TWh in 2025 and remaining below 50 TWh until it increases again after 2045. For most of the time period, the system is really focusing on carbon-free generation. The PV and Wind Off profile like Policy 7a dissipates slowly, probably from curtailment and integration issues, but the Wind On profile is sufficient to sustain compliance. By 2050, CCGT is again dominant with an electricity generation of 242 TWh.

Policy 7b (TWh)	2025	2030	2035	2040	2045	2050
CCGT	42.473	15.700	18.074	46.381	102.398	242.213
PV	20.796	7.819	9.005	4.896	0.000	0.000
Wind Off	12.232	4.926	3.221	2.120	0.000	0.000
Wind ON	47.310	129.138	163.980	177.485	163.332	58.493

Table 70: Electricity generation mix - 24/7 90% Carbon-Free Energy

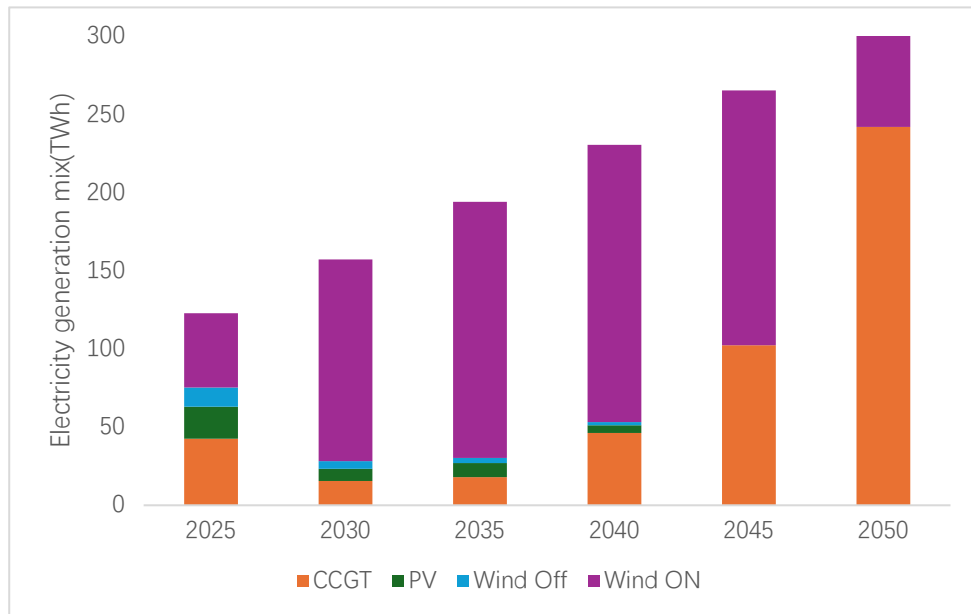


Figure 48: Electricity generation mix - 24/7 90% Carbon-Free Energy

This behavior is evidenced by time-slice level analysis of electricity grid demand. The figure below indicates renewable electricity (light green bars) behaves similarly to total electricity demand (dark green bars) for the majority of the year in 2025. During the spring through autumn period, the system consistently produced close to 90% carbon-free electricity supply per time slice. Overall, this demonstrates that the time-slice constraint was correctly implemented and the system structure for Policy 7b allows for near continuous compliance with a 24/7 carbon-free target.

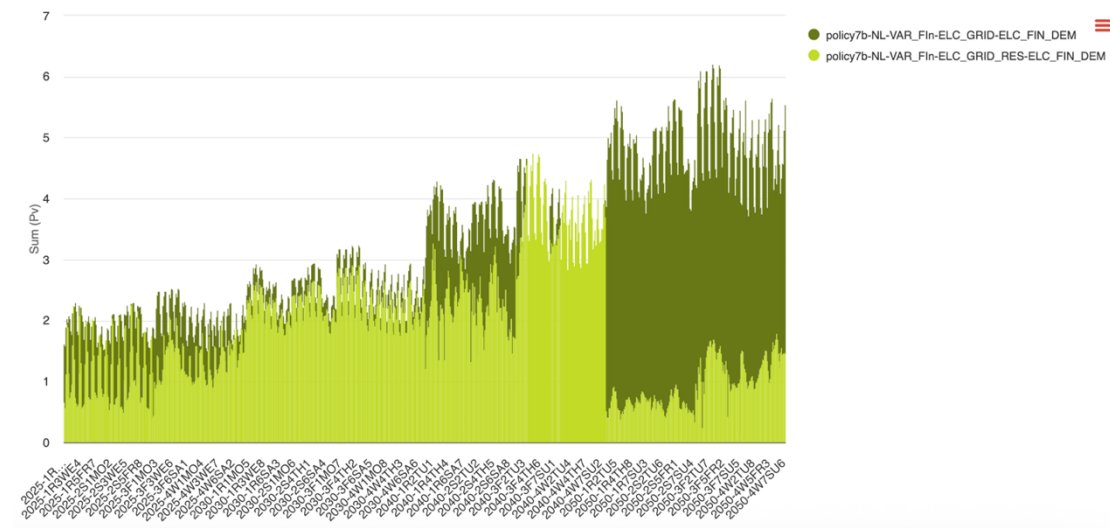


Figure 49: Time-slice level analysis of electricity grid demand - 24/7 90% Carbon-Free Energy

8.2. Overall GHG Emissions

Policy 7B achieves emissions reductions for the early years, emissions drop from 15 Mt in 2025 to about 5.5 Mt in 2030 and remain below 15 Mt until 2040. This reduction is established primarily due to the effective enforcement of the 90% carbon-free constraint. The constraint

ensures that the system acquires a majority of renewable technologies and moves away from fossils.

However, after 2045, emissions begin to rise again, reaching 31.2 Mt by 2045 and 73.8 Mt in 2050. This gap in emission reductions is likely due to the growing reliance on CCGT technologies in the late horizon as the capacity for wind energy becomes saturated and when more flexible low-carbon technologies like storage, hydrogen/CCS, are not available to the system at the same cost as before to keep the cost of constraints low.

Policy 7B (Mtons)	2025	2030	2035	2040	2045	2050
CO2 emission	15.01718	5.551071	6.390475	14.12824	31.19207	73.78189

Table 71: Overall CO2 emissions - 24/7 90% Carbon-Free Energy

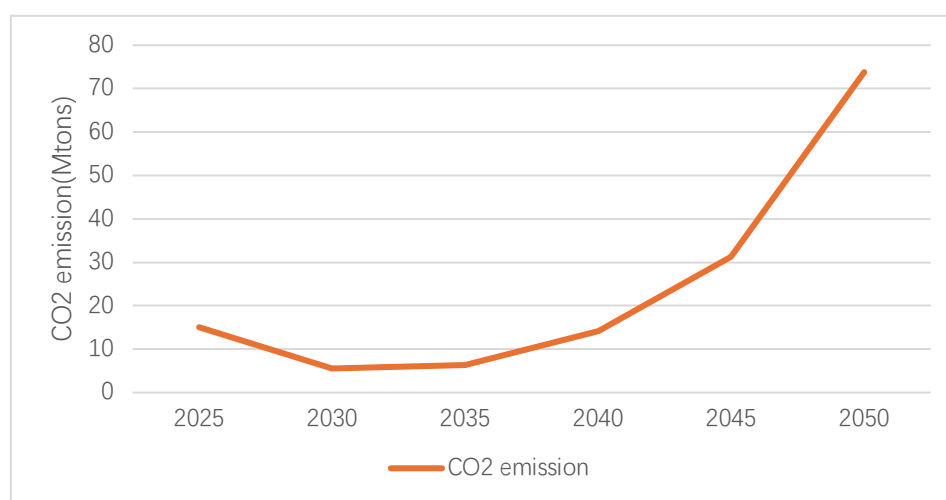


Figure 50: Overall CO2 emissions - 24/7 90% Carbon-Free Energy

Compared to Policy 7A, Policy 7B provides better emissions reductions particularly during the significant mid-term timeframe, but does not achieve length time horizon reductions.

8.3. Carbon Intensity of Electricity Generation

The table below shows a significant drop between 2025 and 2035, from 122.3 kg CO₂/MWh in 2025 to only 33 kg/MWh by 2035. This dramatic decrease is a strong sign of Wind On infiltrating the generation mix and much less reliance on CCGT generation.

By 2040, carbon intensity starts to increase again from 61.4 kg/MWh in 2040 to 245.5 by 2050. This matches with the noticeable increase in gas-fired generation that has returned to the electricity mix by 2040. By 2050, it's nearly twice that of the 2045 value, illustrating how difficult it is to decarbonize the electricity system, given the evident lack of further system flexibility.

This highlights how well Policy 7b performs in the medium term, mostly because of the requirement for 24/7 clean energy, but still has potential challenges for decarbonizing in the longer term.

Policy_7a	2025	2030	2035	2040	2045	2050
CO2 (kg)	15017184.23	5551071.429	6390474.783	14128238.71	31192073.48	73781890.82
Electricity (MWh)	122810.7306	157000	193898.4018	230000	264986.0731	300529.9087
Carbon intensity (kg/MWh)	122.2790888	35.35714286	32.95785176	61.42712482	117.7121239	245.5059836

Table 72: Carbon intensity of electricity generation - 24/7 90% Carbon-Free Energy

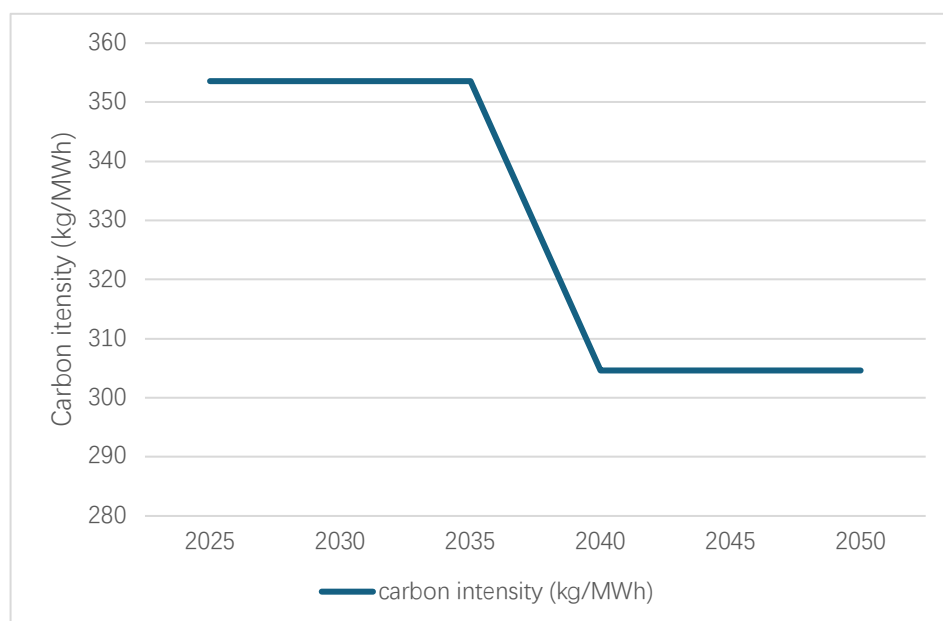


Figure 51: Carbon intensity of electricity generation - 24/7 90% Carbon-Free Energy

8.4. New Electrical Capacities

The spending levels increase to very high levels on onshore wind (ELE_NEW_WIND_ON), particularly in the early years. By 2030, about 60.5 GW of new wind capacity has been installed, following the addition of 12.8 GW in 2025. This very early and rapid growth gives the system time to provide the 90% carbon-free across the time slices.

On the other hand, the new CCGT capacity (ELE_NEW_CCGT) has small capacity additions across the time horizon. There are minor additions in 2040 (12.98 GW) and 2045 (7.48 GW), which would suggest that fossil capacity is only installed when it is absolutely necessary. Then in 2050, there is another small addition (12.99 GW), which may reflect some increasing strain on the system as the Wind On capacity begins to level.

Policy 7B (GW)		2025	2030	2035	2040	2045	2050
ELE_NEW_CCGT	VAR_Ncap				12.98	7.48	12.99
ELE_NEW_WIND_ON	VAR_Ncap	12.79	60.54				22.63

Table 73: New electrical capacities - 24/7 90% Carbon-Free Energy

Data from the graph below show, that under Policy 7b, most financing is invested in renewables first, while gas is minimized. This is consistent with decarbonization and of a focus on clean generation technologies.

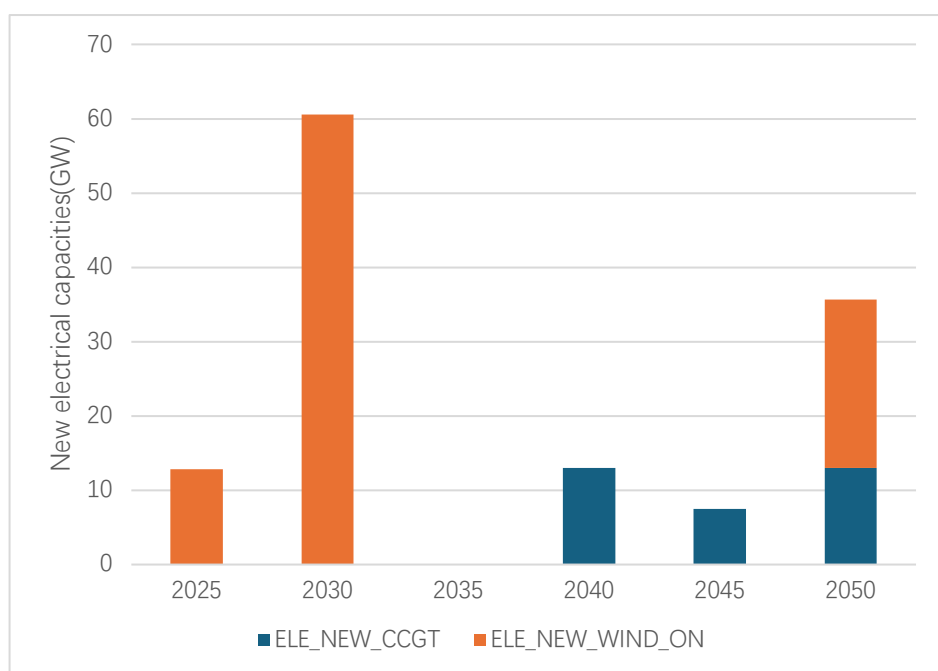


Figure 52: New electrical capacities - 24/7 90% Carbon-Free Energy

8.5. Full Load Operation Hours of CCGT

In the years between 2025 and 2035, FLH drastically reduce, reaching their lowest point of 652 h/year in 2030 (a noticeable reduction from 1,766h/year in 2025) due to growth in Wind On capacity and consequent shift from Fossil-based generation. During this period, CCGT plants are mostly used only as necessary under the 24/7 constraint.

From 2040 onwards, FLH start increasing rapidly, then peaking at 5,000h in 2045 and at over 7,200h in 2050, indicating CCGT is being used effectively as baseload. This demonstrates without further investment in firm carbon-free capacity or storage, the model ultimately resists moving past fossil fuels and returns to its reliance on fossil fuels to manage supply reliability (particularly since renewable additions begin to level off in the last decade).

Policy_7b	2025	2030	2035	2040	2045	2050
Annual activity (PJ)	152.9	56.51	65.066	166.97	368.63	871.96
Capacity (GW)	24.044	24.044	24.044	12.98	20.46	33.46
Full load hours	1766.437457	652.854	751.7006	3573.232	5004.752	7238.826

Table 74: Full load operation hours of CCGT - 24/7 90% Carbon-Free Energy

The full-load hours (FLH) of CCGT under Policy 7b have an almost U-shaped pattern, showing its strategic lower position in the earlier years and ultimately its dominant role in the end year.

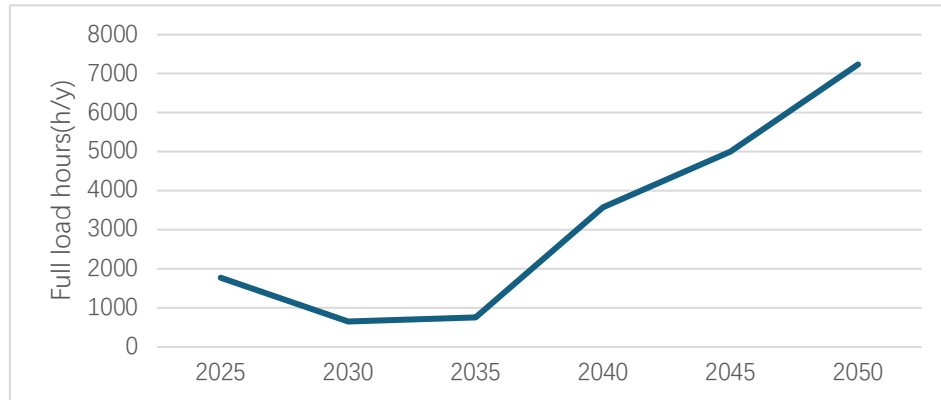


Figure 53: Full load operation hours of CCGT - 24/7 90% Carbon-Free Energy

8.6. RES Curtailment

Curtailment is highest in 2030 under the 90% carbon-free case at 51.7% for PV, 47.6% for offshore wind, and 36.6% for onshore wind. This high values are caused by overproduction or grid constraints during fast RES extension. Curtailment reduces gradually in the following years and dropped to 0–2% by 2045, indicating better system balancing. However, the 90% target leads to higher overall curtailment, in particular in the short and medium-term.

Policy_7b	2025	2030	2035	2040	2045	2050
PV (TWh)	0.785068	8.36941	1.780299	0.49629	0	0
PV (%)	3.64%	51.70%	16.51%	9.20%	0%	0%
Wind OFF (TWh)	0.301564	4.473941	3.046296	1.013341	0	0
Wind OFF (%)	2.41%	47.59%	48.61%	32.34%	0%	0%
Wind ON total (TWh)	1.071905	74.66299	36.11636	18.90555	3.546483	0.979054
Wind ON (%)	2.22%	36.64%	18.05%	9.63%	2.13%	1.65%

Table 75: Renewable Energy Sources Curtailment - 24/7 90% Carbon-Free Energy

8.7. RES Share in Final Electricity Demand

The RES share peaks at 95.1% of final electricity demand in 2030, and it remains fairly high in 2035, indicating the significant contribution of renewables in meeting the ambitious requirement of 90% carbon-free energy in each time slice, particularly in the medium term.

From 2040, the RES share declines progressively starting with 59.8% in 2040, 44.4% in 2045, and just 17.4% in 2050 and this decline is associated with:

- the lessened amount of PV and offshore wind;
- the levelling of Wind On capacity; and
- increased reliance on gas (CCGT) which is used to ensure system reliability.

	2025	2030	2035	2040	2045	2050
policy_7b	58.50%	95.11%	78.10%	59.75%	44.38%	17.37%

Table 76: Renewable Energy Sources share in final electricity demand - 24/7 90% Carbon-Free Energy

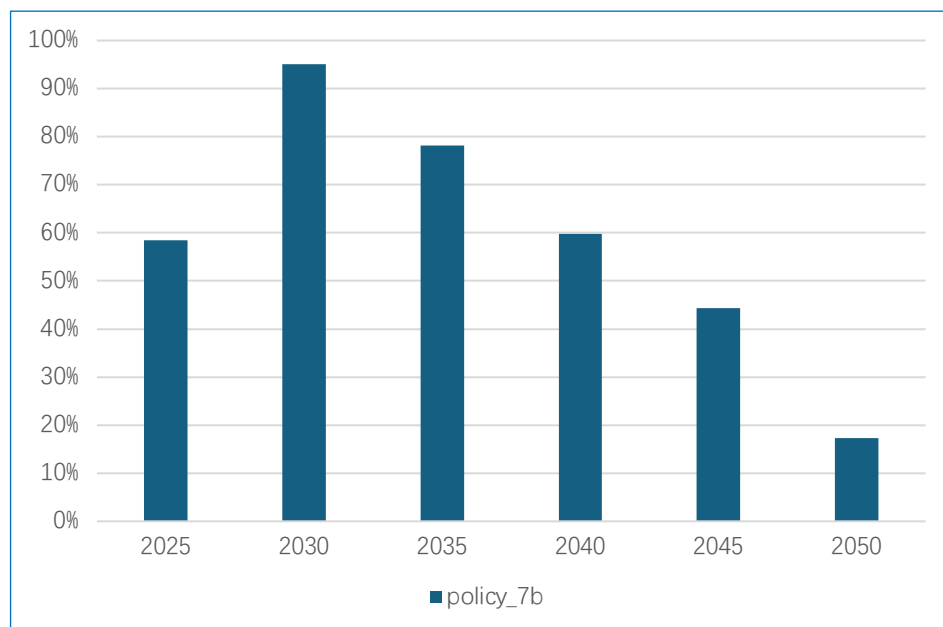


Figure 54: Renewable Energy Sources share in final electricity demand - 24/7 90% Carbon-Free Energy

8.8. RES Share Results on the Time-slice Level

The time-slice-level results for Policy 7b reveal a significant temporal mismatch of renewable electricity supply against its demand. The light blue line (VAR_FIn from RES to grid) is the renewable electricity actually delivered, and green line (VAR_FIn from total grid) is total electricity delivered to the grid.

In the early years (2025-2035), renewable electricity is closely related to total grid delivery, in fact it frequently equals or exceeds total electricity delivery in some time-slices. This indicates that the 90% carbon-free requirement is likely enforced at the time-slice level, particularly from years 2030-2035 when RES penetration is at its highest levels.

From 2045 onward, though, a difference can be seen: the gap between total electricity demand and renewable delivery is getting larger, particularly in the later years. This indicates a reliance

on non-renewable sources (namely CCGT) to cover the gap to meet system demand. Reaching 2050, the renewable share has dramatically reduced in most time-slices confirming that without further investments in flexibility or storage, the system cannot operate to the original 90% clean energy targets anymore.

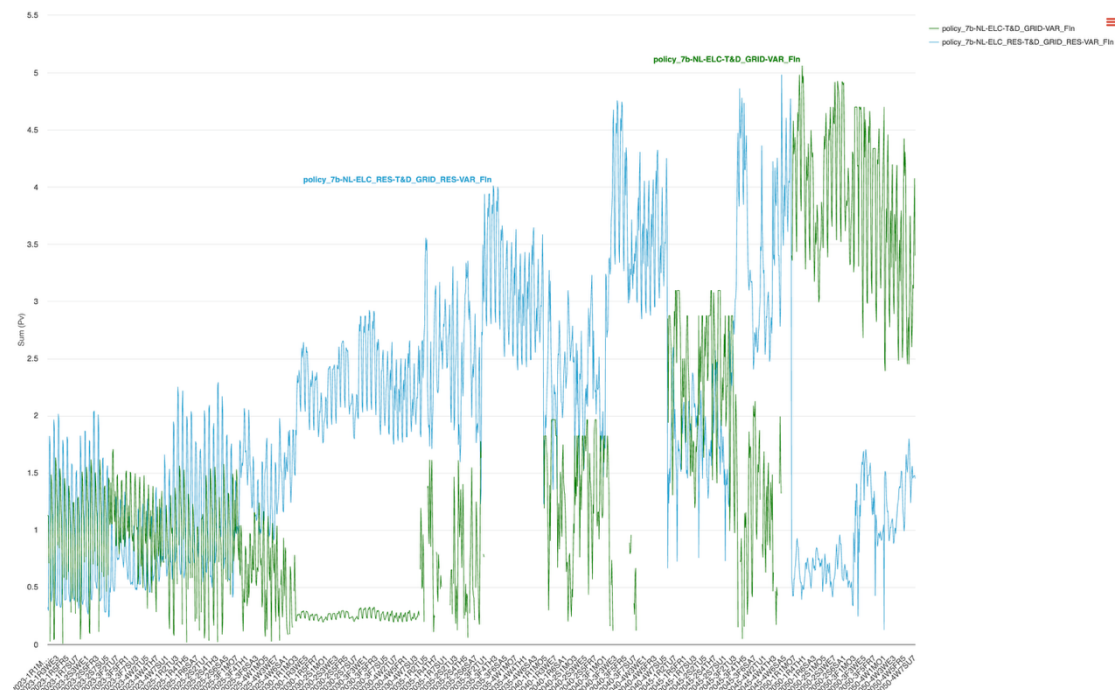


Figure 55: Renewable Energy Sources share results on the time-slice level - 24/7 90% Carbon-Free Energy

8.9. Electricity Generation Cost

Costs for electricity generation under this policy change across technologies and times. In the case of CCGT, the costs start relatively low in 2025 (€18.5/MWh) and increase progressively by 2030 and 2035 (up to €71.5/MWh) due to use hovering around zero. After 2040 usage increases and the costs decline to €20.40/MWh by 2050. This suggests that there are high fixed costs distributed over a small amount of generation initially and then more efficiently used later on.

PV and Wind Off have low costs initially (PV: 9.9 €/MWh & Wind Off: 2.4 €/MWh), but are then phased out entirely after 2040, most likely due to high curtailment or limits to integrating their output into the system.

Lastly, Wind On is the most expensive source in all years, costing almost €90/MWh in 2030, with costs down to €60/MWh by 2050. These higher costs, as compared to other technologies examined, are expected to reflect high capital costs, and perhaps, overcapacity, as generation exceed demand and curtailment occurs.

Policy7b	Item	2025	2030	2035	2040	2045	2050
CCGT	Cost (MEuro)	786.680	967.320	1291.660	1571.130	2687.240	4932.200
	Electricity (TWh)	42.473	15.700	18.074	46.381	102.398	242.213
	Electricity cost (Euro/MWh)	18.522	61.613	71.465	33.875	26.243	20.363
PV	Cost (MEuro)	205.360	154.022	102.680	51.340	0.000	0.000
	Electricity (TWh)	20.796	7.819	9.005	4.896	0.000	0.000
	Electricity cost (Euro/MWh)	9.875	19.698	11.403	10.485	0.000	0.000
Wind off	Cost (MEuro)	29.270	21.950	14.630	7.310	0.000	0.000
	Electricity (TWh)	12.232	4.926	3.221	2.120	0.000	0.000
	Electricity cost (Euro/MWh)	2.393	4.456	4.543	3.448	0.000	0.000
Wind on	Cost (MEuro)	1064.680	11556.220	11542.150	11528.088	9505.600	3553.630
	Electricity (TWh)	47.310	129.138	163.980	177.485	163.332	58.493
	Electricity cost (Euro/MWh)	22.505	89.487	70.388	64.953	58.198	60.754

Table 77: Electricity generation cost - 24/7 90% Carbon-Free Energy

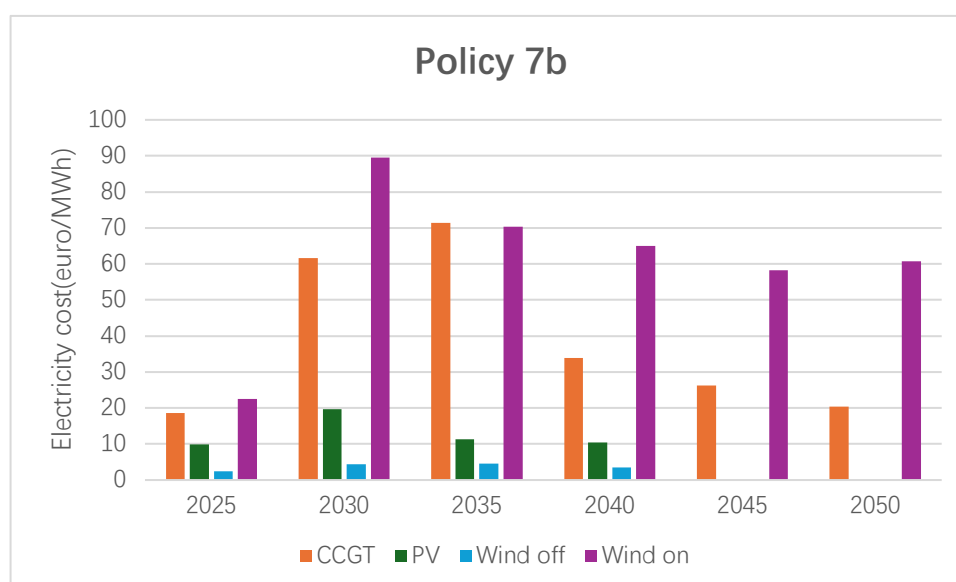


Figure 56: Electricity generation cost - 24/7 90% Carbon-Free Energy

Cross Comparison

In this chapter, we select key criteria for cross-comparison to assess which policies are the most effective in controlling greenhouse gas (GHG) emissions. The selected criteria are:

- **Overall GHG emissions:** This represents the total amount of CO₂ emitted by the energy system, in Mton.
- **Carbon intensity:** This indicates the amount of GHG emissions produced per unit of electricity generated, reflecting the cleanliness of the electricity mix, in kg/MWh.

CO2 emissions (Mtons)	2025	2030	2035	2040	2045	2050
0.Baseline	15.017	23.888	39.816	48.714	64.706	73.385
1. Carbon tax	10.717	8.141	2.060	0.000	0.000	0.000
2.Limit CO2	15.017	16.000	16.000	16.000	16.000	16.000
3.Imfst	14.550	22.374	31.896	35.377	51.330	63.826
4.Subsidies	8.783	7.100	7.273	6.424	7.876	8.932
5. Recycling carbon tax	8.657	6.402	1.364	0.000	0.000	0.000
6.RES50%	15.017	27.552	33.502	33.757	38.685	43.874
7a. Carbon free 50%	15.017	27.755	34.227	44.283	65.062	73.837
7b. Carbon free90%	15.017	5.551	6.390	14.128	31.192	73.782

Table 78: Cross Comparison of Overall Carbon emissions

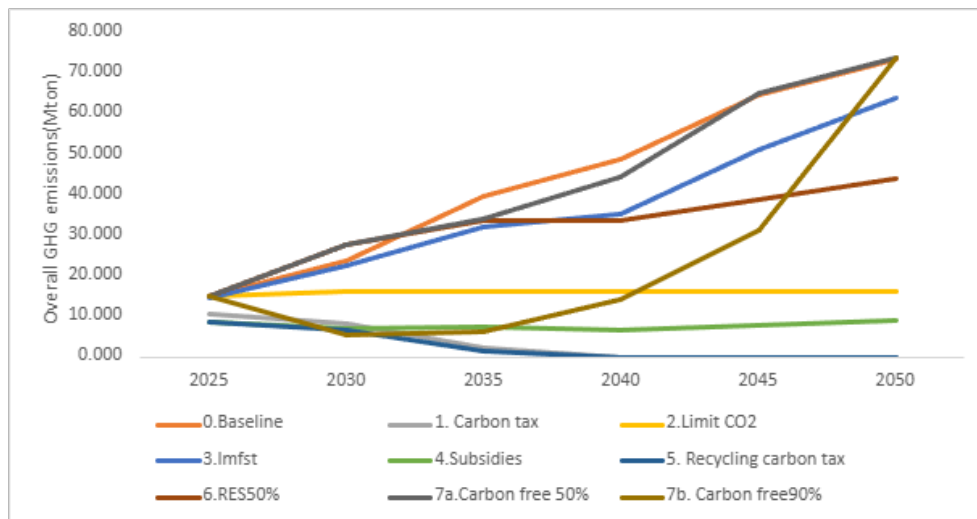


Figure 57: Cross Comparison of Overall Carbon emissions

From the figure, it can be seen that policies recycling carbon tax, carbon tax, subsidies and limiting CO2 are the top 4 most effective policies applied in controlling the overall GHG emissions.

Carbon intensity (kg/MWh)	2025	2030	2035	2040	2045	2050
0.Baseline	122.281	152.152	205.342	211.799	244.187	244.186
1. Carbon tax	87.265	51.851	10.625	0.000	0.000	0.000
2.Limit CO2	122.281	101.911	82.517	69.565	60.381	53.240
3.Imfst	118.48	142.51	164.50	153.81	193.71	212.38
4.Subsidies	71.515	45.220	37.510	27.929	29.721	29.721
5. Recycling carbon tax	67.012	36.377	5.901	0.000	0.000	0.000
6.RES50%	122.281	175.488	172.713	146.716	145.940	145.939
7a.Carbon free 50%	122.279	176.786	176.518	192.536	245.532	245.689
7b. Carbon free90%	353.571	353.571	353.571	304.615	304.615	304.615
Combination	22.687	14.717	0.044	0.000	0.000	0.000

Table 79: Cross Comparison of Overall Carbon emissions

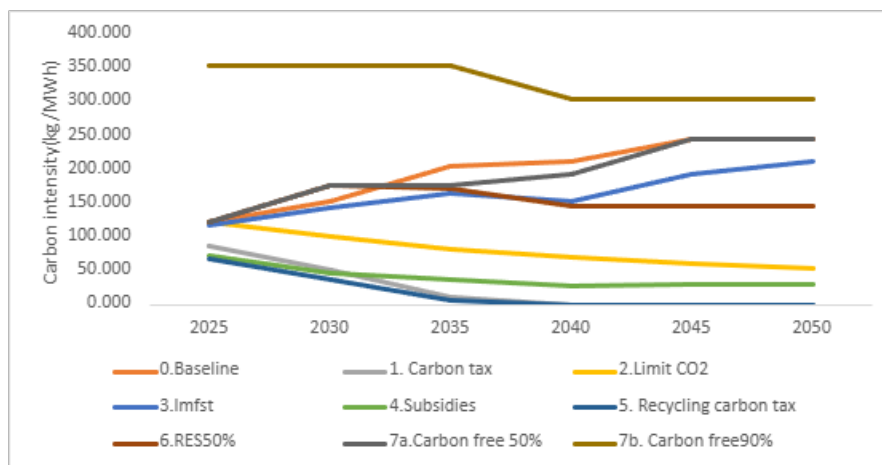


Figure 58: Cross Comparison of Carbon intensity

From the figure, it can be seen that recycling carbon tax, carbon tax, subsidies and CO2 cap are the most effective policies in decreasing carbon intensity. This result matches the results in overall GHG analysis.

Result Analysis of Policy Combination

A further analysis of policy combination is conducted based on the cross-comparison results. Top 4 policies are set together to test its effectiveness.

Top 4 Policies	Overall GHG	Carbon Intensity	Policy Combination
1	Recycling carbon tax	Recycling carbon tax	- Recycling carbon tax (includes carbon tax) - Subsidies - CO2 Cap
2	Carbon tax	Carbon tax	
3	Subsidies	Subsidies	
4	CO2 cap	CO2 cap	

Table 80: Top Effective Policies in Environmental Sustainability

A Policies combination of recycling carbon tax, subsidies and CO2 cap is run to further analysis the effect of Signal policy vs. Policy combination. The results are listed as follows.

Policy Combination	2025	2030	2035	2040	2045	2050
CO2 emissions (Mtons)	3.256	2.723	0.010	0	0	0
Carbon intensity (kg/MWh)	22.687	14.717	0.044	0	0	0

Table 81: CO2 emissions and carbon intensity from Policy Combination Case

From the table, it can be seen by applying the policy combination of subsidies, carbon tax, CO2 cap and Recycling carbon tax, CO2 emissions and Carbon intensity can achieve zero from year 2040. To further understand how this is realized, energy share is analyzed in the following table:

Combination	2025	2030	2035	2040	2045	2050
RES	133.853	176.383	224.417	271.100	312.728	354.675
CCGT	9.208	7.703	0.028	0.000	0.000	0.000
Import	0.456	0.969	8.939	10.478	13.722	15.564
Grid demand	143.517	185.058	233.386	281.581	326.450	370.239
RES Share	93.27%	95.31%	96.16%	96.28%	95.80%	95.80%

Table 82: Energy generation share of Policy Combination Case

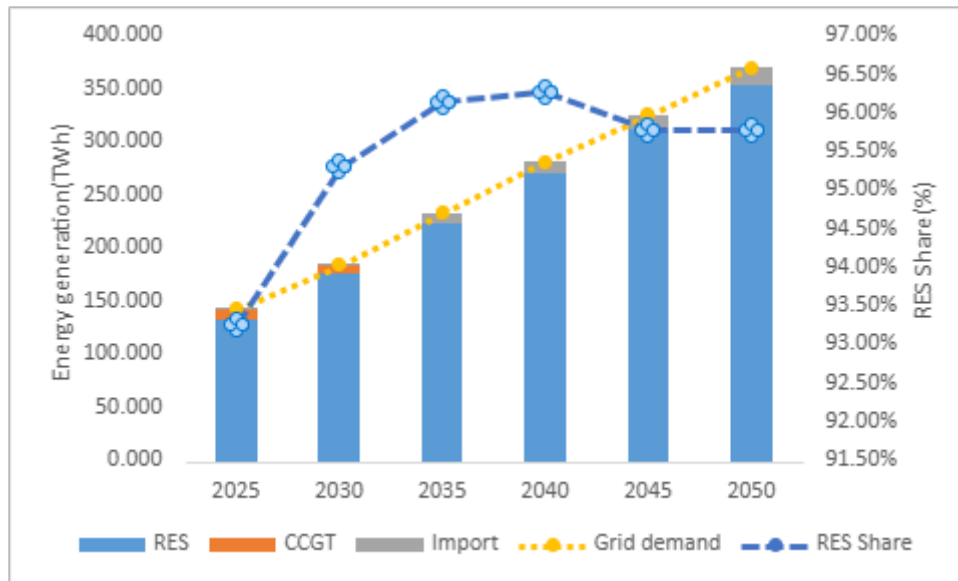


Figure 59: Energy generation share of Policy Combination Case

As checked, from 2040 all CCGT is abandoned, the whole country's electricity is provided by over 95% RES, and the rest is imported. Which explains why CO₂ emissions are 0. As can be concluded, this combination of Policy is very effective in achieving zero carbon emission target by 2050.

Conclusion

This report analyzed the effectiveness of diverse policy instruments in shaping the Dutch electricity system to be more sustainable and low carbon. This study assessed individual and combined policies on greenhouse gas (GHG) emissions, carbon intensity, renewable energy integration, and system costs, through detailed scenario-based analysis between 2025 and 2050 using the TIMES model.

From the individual policies, the recycling carbon tax, the carbon tax, capital subsidies, and CO₂ emissions cap were the most effective policies with respect to reducing the overall GHG emissions and carbon intensity associated with power demand. These measures effectively advanced the transition from fossil fuel, including CCGT, and encouraged the development of renewable technology, including onshore wind.

The cross-comparisons provided evidence that:

- Carbon tax earned with recycling resulted in the lowest carbon intensity and the greatest emissions reduction in the early stages.
- CO₂ cap imposed a strict maximum for emissions which kept the emissions trajectory flat.
- With capital subsidies, investments were pushed into offshore wind with a high proportion of renewables.
- Carbon tax substantially reduced the use of fossil fuel and expanded RES.

These top four policy contexts combined have the most substantial impact. Starting in 2040, the combination resulted in:

- Zero CO₂ emissions and zero carbon intensity, achieving a complete decarbonization of the power system.
- More than 95% RES, with full phase-out of CCGT, and the remaining demand covered by imports.
- A RES electricity mix with high stability, variety and safety for rising and new electricity demand purposes.

These findings validate that total decarbonization cannot be met by just implementing one policy. A comprehensive analysis and selection, that combines price-based tools (like a carbon tax), regulatory restrictions (like CO₂ caps) and supportive policies (like subsidies and revenue recycling) is needed to accelerate the energy transition while preserving affordability and reliability.

Overall, this study demonstrates that policy combination is not only more effective but also crucial for achieving deep decarbonization, demonstrating that ambitious and integrated action can lead a carbon-free power systems by 2050.